

Lierda NT26-KCN E Series Hardware Design Manual

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Document Revision History

Document Version	Change Date	Reviser	Reviewer	Change content
Rev1.0	26-04-17	CWY	-	Initial version
Rev1.1	26-04-23	CWY	-	Add description of NT26K2E1-0G module.

Safety Instructions

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Road safety comes first! When driving, do not use handheld mobile devices unless they have a hands-free function. Please park before making a call!



Please turn off mobile terminal devices before boarding. The wireless functions of mobile terminals are prohibited from being activated on the plane to prevent interference with the aircraft's communication system. Ignoring this instruction may affect flight safety and even violate the law.



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Your mobile terminal device will receive and transmit radio frequency signals when powered on, which may cause radio frequency interference when near televisions, radios, computers, or other electronic devices.



Keep mobile terminal devices away from flammable gases. When near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off mobile terminal devices. Operating electronic devices in any potentially explosive areas poses safety risks.

Applicable Module Selection

Serial Number	Module model	Supported frequency bands	Instructions
1	NT26K2E0-0G	Band1/3/5/8/34/38/39/40/41*	VDD_EXT power down in sleep mode
2	NT26K2E1-0G	Band1/3/5/8/34/38/39/40/41*	VDD_EXT power down in sleep mode, the module only supports single Beidou version.

Precautions

LTE-TDD B41 only supports 140M (2535-2675MHz).

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1 Introduction

This document defines the application specifications of Lierda NT26-KCN E series Cat.1bis wireless communication module, describing its frequency band functions, key features, hardware interfaces, operating modes, electrical characteristics, mechanical specifications, as well as packaging and storage. It helps users quickly grasp the application methods of this Cat.1bis module and enables quick and flexible product design.

1.1 Special symbols

Table 1-1 Special Symbols Description

Symbol	Definition
*	If not specifically stated, the asterisk (*) marked after the module functions, features, interfaces, pin names, AT commands, or parameters indicates that these module functions, features, interfaces, pin names, AT commands, or parameters are under development and currently not supported.

2 Product Overview

The NT26-KCN E series module is a globally leading Cat.1bis wireless communication module, compliant with the 3GPP R14 Cat.1bis standard. It supports bandwidths of 1.4/3/5/10/15/20MHz and features a compact size, low power consumption, and strong anti-interference capabilities. The module is suitable for various common IoT application scenarios, such as:

- ◆ Smart Meter Reading
- ◆ Intelligent Parking
- ◆ Smart City
- ◆ Intelligent Security
- ◆ Asset Tracking
- ◆ Smart Home Appliances
- ◆ Agriculture and environmental monitoring, etc.

2.1 Key Features

The key features of the NT26-KCN E series module are as follows:

Table 2-2 Description of Key Module Features

Parameter	Explanation
Module packaging	LCC+LGA
Module dimensions	17.7mm×15.8mm×2.4mm (L×W×H)
Module weight	1.3g
Operating voltage	◆ Power supply voltage: 3.3~4.5V, typical value: 3.8V ◆ Extended power supply voltage(1): 3.1~4.5V
VDD_EXT characteristics	The NT26-KCN E series offers two hardware versions. Sleep mode: VDD_EXT power down Sleep mode: VDD_EXT does not power down
Ultra-low power consumption	Shutdown mode: $\leq 2 \mu A$

Transmission power	23dBm±2.7dB (Max)
Antenna interface	50Ω characteristic impedance
USIM	Supported USIM card types: Class B (3.0 V) and Class C (1.8 V), see section 4.3 for details.
UART	Main Serial Port 1: ◆ Used for AT command transmission and data transfer, the supported baud rate defaults to 115200bps. ◆ Used for firmware upgrade, the supported baud rate defaults to 921600bps.
	Debugging serial port 1: ◆ Used for software Debug debugging ◆ The default baud rate of the serial port is 3 Mbps.
	Auxiliary Serial Port 1: ◆ User-defined function
USB	◆ 1 USB port ◆ Compatible with USB 2.0 (supports slave mode only), maximum data transfer rate of 480 Mbps ◆ Used for AT command transmission, data transfer, software debugging, and firmware upgrade
I2C	◆ 1-way I2C interface ◆ Compliant with I2C bus protocol specifications ◆ Supports standard mode 100KHz, fast mode 400KHz
SPK	◆ 1-channel analog audio interface: ◆ Used for audio, can directly drive low-power headphones or connect to an external PA to drive speakers.
Indicator light	◆ Operating Status Indication ◆ Network Status Indicator
Communication interface characteristics	Support 3GPP Rel.13/14 Cat.1 radio communication interface and protocol
Network Protocol Features (2)	Supports TCP/UDP/HTTP/SSL*/MQTT/FOTA/PPP*/RNDIS*/FTP*/ECM* and other protocols.
Data Transmission	◆ Maximum support for Cat.1 FDD and TDD

Characteristics	◆ 1.4/3/5/10/15/20MHz RF bandwidth ◆ Uplink QPSK, 16QAM ◆ Downlink QPSK, 16QAM, 64QAM ◆ LTE-FDD: Maximum downlink rate 10Mbps, maximum uplink rate 5Mbps ◆ LTE-TDD: Maximum downlink rate 8.96Mbps, maximum uplink rate 3.1Mbps
Firmware upgrade	◆ Main Serial Port Upgrade ◆ USB interface upgrade ◆ DFOTA Upgrade
Operating temperature	◆ Normal operating temperature range (3): -35~75°C ◆ Extended operating temperature range (4): -40~85°C
Storage temperature	Storage temperature range (5): -40~90°C
RoHS	All components comply with EU RoHS standards.

Remarks

(1)For a detailed description of the extended voltage, see Chapter 6.2.

(2) The network protocols supported by different firmware versions may vary, subject to the description of the specific firmware version in use.

(3)(4)(5) For specific descriptions of temperature, see Chapter 6.6.

2.2 Functional block diagram

The NT26-KCN E series module features rich peripheral interfaces and RF functionality, with the functional block diagram as follows:

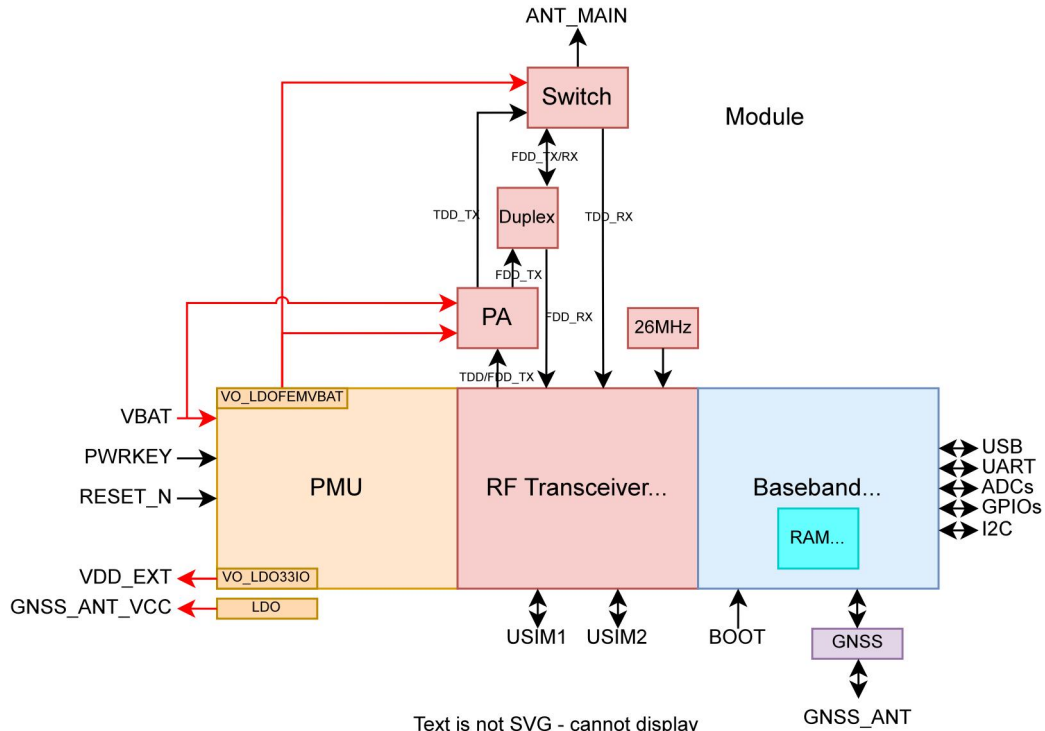


Figure 2.1 Module Function Block Diagram

2.3 Pin distribution diagram

The NT26-KCN E series module has a total of 109 pins, of which 48 are LCC pins and the remaining 61 are LGA pins. The module pin distribution diagram is as follows:

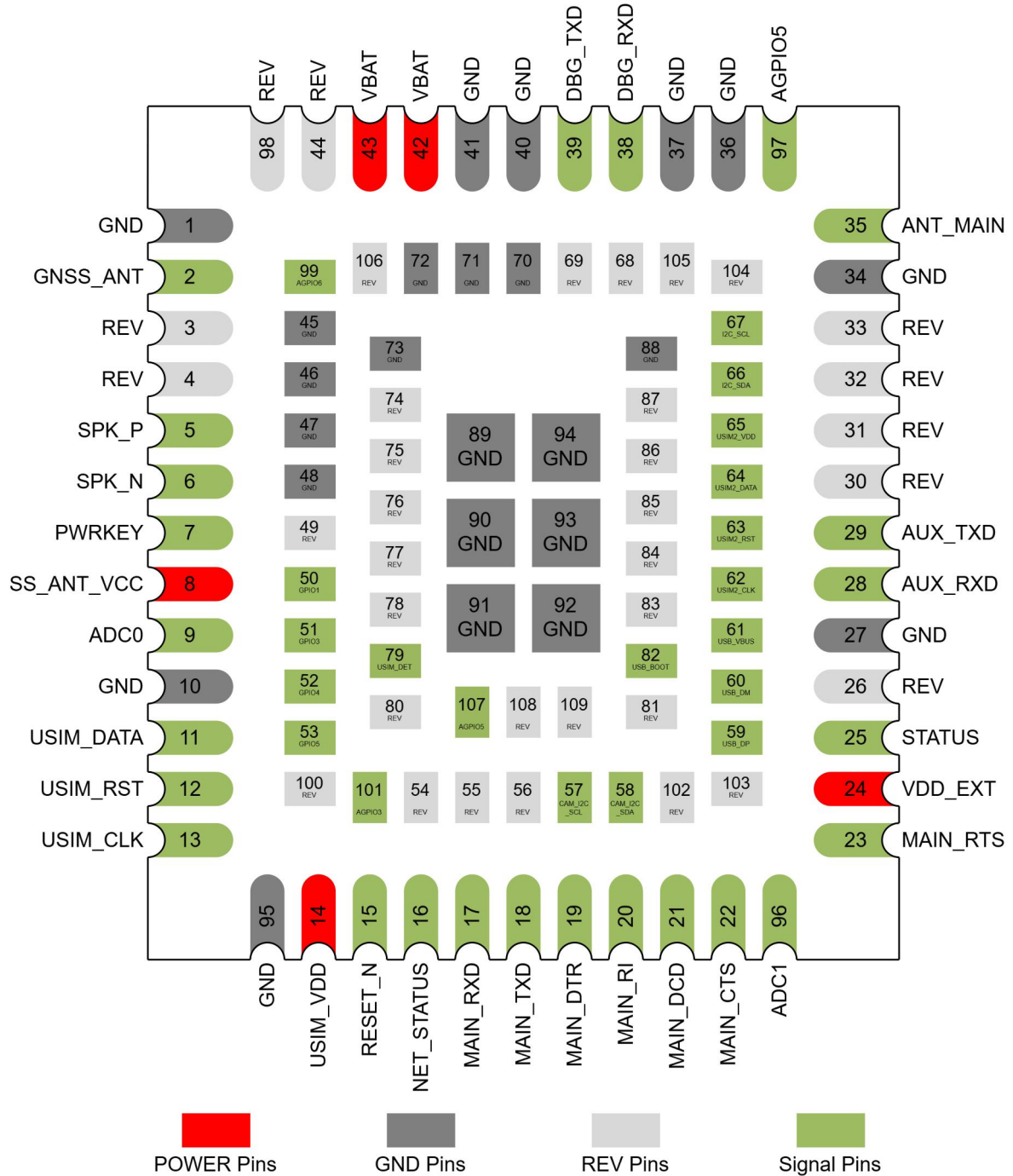


Figure 2.2 Module Pin Assignment Diagram

2.4 Pin description table

To better understand the application, the following table provides type definitions for I/O parameters:

Table 2-4 Type Definition Description of I/O Parameters

I/O parameter types	Explanation
DI	Digital input
DO	Digital output
DIO	Digital Input and Output
AI	Simulated input
AO	Simulated output
AIO	Simulated Input and Output
PI	Power Input
PO	Power output
G	Grounding
OD	Open Drain

To better understand the application, the following table provides a description of the characteristics of the power domain parameters:

Table 2-5 Module Power Domain Characteristics Description

Power domain parameters Type	DC characteristics	Explanation	Power supply object
VBAT	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	Module power input, it is recommended to use a 3.8V/1.2A power supply.	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 1.8V; Different hardware versions: In sleep mode, VDD_EXT power-down version, drive capability 120mA; In sleep mode, VDD_EXT does not power down, drive capability 3mA.	VDD_EXT
VO_LDOIO	VOLmax=0.2×VO_LDOIO VOHmin=0.7×VO_LDOIO VILmax=0.15×VO_LDOIO VIHmin=0.8×VO_LDOIO	Default 1.8V, software configurable; Power loss in sleep mode	UART GPIO

VDD18AON	VILmax=0.2×VDD18AON VIHmin =0.7×VDD18AON VOLmax=0.15×VDD18AON VOHmin=0.8×VDD18AON	No power loss in sleep mode.	WAKEUP RESET
LDO_AONIO	VILmax=0.2×LDO_AONIO VIHmin =0.7×LDO_AONIO VOLmax=0.15×LDO_AONIO VOHmin=0.8×LDO_AONIO	Default 1.8V, software configurable; No power loss in sleep mode.	AGPIO
VO_LDOSIM	Vnorm=1.8/3.0V VOLmax=0.15×VO_LDOSIM VOHmin=0.8×VO_LDOSIM VILmax=0.2×VO_LDOSIM VIHmin=0.7×VO_LDOSIM	Dedicated SIM card power supply, supports 1.8/3.0V cards.	USIM
GNSS_ANT_VCC	Vnorm=3.3V	GNSS active antenna power supply, maximum driving capability: 200mA	Use of GNSS Active Antenna

Remarks

◆ Power domain: An area in a chip or system that is divided for independent power supply, where each area can independently control the power on/off, voltage levels, or power consumption modes.

◆ VDD_EXT and VO_LDOIO share the same power supply.

Table 2-6 Module Pin Description

1. Power Supply POWER					
Pin number	Pin name	I/O	Description	DC characteristics	Remarks
42,43	VBAT	PI	Power supply	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	The external power supply needs to provide a current-carrying capacity of at least 1.2A.
24	VDD_EXT	PO	Output power supply	Vnorm=1.8V I _o max=120mA/3mA	Can be used for external circuit pull-up or reference level; in sleep

					mode, there are two modes: power-down and non-power-down; if not used, leave it floating, it is recommended to reserve a test point.
8	GNSS_ANT_VCC	PO	Output power source	V _{norm} =3.3V I _o max=200mA	GNSS antenna power supply
1,10,27,34, 36~37,40~41,45~48, 70~72,73,88~95	GND	G	Grounding	-	-

2. Power On/Off and Reset

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
7	PWRKEY	DI	Module power on/off pin	V _{IL} max=0.5V	Internal pull-up, active low.
15	RESET_N	DI	Module reset pin	V _{IL} max=0.5V	Internal pull-up, active low.

3、USB

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
59	USB_DP*	DIO	USB differential data (+)	-	Complies with USB 2.0 specifications. Requires 90 Ω differential impedance. Test points need to be reserved. Leave them floating if not used.
60	USB_DM*	DIO	USB differential data (-)	-	
61	USB_VBUS*	DI	USB_VBUS input wake-up	V _{max} = 5.25 V V _{min} = 3.0 V	

				Vnom = 5.0 V	(Default is off)
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4. Serial Port μ ART					
Pin number	Pin name	I/O	Description	DC characteristics	Remarks
17	MAIN_RXD	DI	The main serial port receives data.	VO_LDOIO	If not used, leave floating
18	MAIN_TXD	DO	The main serial port sends data.	VO_LDOIO	If not used, leave floating
19	MAIN_DTR	DI	The main serial port data terminal is ready.	VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.
20	MAIN_RI	DO	The main serial port outputs a ringing prompt.	LDO_AONIO /VDD18AON	Default high level, when the module has URC to report, it outputs a low level for 120ms, and after the data output is completed, it returns to a high level.
21	MAIN_DCD*	DO	The main serial port outputs carrier detection.	VO_LDOIO	Internally connected in parallel with pin 66, cannot be used simultaneously.
22	MAIN_CTS*	DO	The main serial port DTE clears sending.	VO_LDOIO	Internally connected in parallel with pin 57, cannot be used

					simultaneously.
23	MAIN_RTS*	DI	The main serial port DTE requests to send.	VO_LDOIO	Internally paralleled with pin 58, cannot be used simultaneously.
38	DBG_RXD	DI	Debug serial port receiving data	VO_LDOIO	If not used, leave floating
39	DBG_TXD	DO	Debugging serial port data transmission	VO_LDOIO	If not used, leave floating
28	AUX_RXD	DI	Assist serial port in receiving data	VO_LDOIO	Internal GNSS use only, external use prohibited. Internal parallel connection with pin 52, cannot be used simultaneously.
29	AUX_TXD	DO	Assist serial port in sending data.	VO_LDOIO	Internal GNSS use only, external use prohibited; Internally connected in parallel with pin 53, cannot be used simultaneously.

5. External USIM Card Interface

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
11	USIM_DATA	DIO	SIM card data cable	VO_LDOSIM	
12	USIM_RST	DIO	SIM card reset line	VO_LDOSIM	
13	USIM_CLK	DIO	SIM card clock line	VO_LDOSIM	
14	USIM_VDD	PO	SIM card power supply	VO_LDOSIM	1.8/3.0V
62	USIM2_CLK	DIO	SIM2 (secondary card) clock line	VO_LDOIO	

63	USIM2_RST	DIO	SIM2 (secondary card) reset line	VO_LDOIO	Internal connection in parallel with pin 50, cannot be used simultaneously.
64	USIM2_DATA	DIO	SIM2 (secondary card) data cable	VO_LDOIO	Internally connected in parallel with pin 51, cannot be used simultaneously.
65	USIM2_VDD	PO	SIM2 (secondary card) power supply	VO_LDOSIM	The same power supply as USIM_VDD
79	USIM_DET	DI	SIM card status signal detection	VDD18AON	Active High

6. Antenna Interface RF

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
35	ANT_MAIN	AIO	RF antenna interface	-	50Ω characteristic impedance
2	GNSS_ANT	AIO	GNSS antenna interface	-	50Ω characteristic impedance

7. I2C Interface

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
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57	CAM_I2C_SCL	DO	I2C Serial Clock	VO_LDOIO	Internally connected in parallel with pin 22, cannot be used simultaneously.
58	CAM_I2C_SDA	DIO	I2C serial data	VO_LDOIO	Internally connected in parallel with pin 23, cannot be used simultaneously.
66	I2C_SDA	DIO	I2C serial data	VO_LDOIO	Internally connected in parallel with pin 21, cannot be used simultaneously; when used as an I2C data signal, an external 4.7K pull-up resistor is required, otherwise leave it floating.
67	I2C_SCL	DO	I2C Serial Clock	VO_LDOIO	The I2C clock signal requires an external 4.7K pull-up resistor; otherwise, it will be left floating.

8. Analog Audio Interface*

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
5	SPK_P	DO	Audio positive differential output	VO_LDOIO	Internal and pin 82 are connected in parallel and cannot be used simultaneously.
6	SPK_N	DO	Audio negative differential output	Internal grounding	If not used, leave floating

9. ADC Interface

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
9	ADC0	AI	Universal ADC Interface	Voltage range: 0~1.05V	If not used, leave floating
96	ADC1	AI	General ADC Interface		If not used, leave floating

10. Status Indication

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
16	NET_STATUS	DO	Network status indication	VDD18AON	If not used, leave floating
25	STATUS	DO	Operating status indication	LDO_AONIO /VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no

					power loss during sleep.
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11. Other Interfaces

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
82	USB_BOOT	DI	Emergency download mode control. Before the module is powered on, pull up to VDD_EXT, and the module will enter emergency download mode after startup.	VO_LDOIO	Internal connection with pin 5 in parallel, cannot be used simultaneously. The module has a default internal weak pull-down, active high. This pin is prohibited from being pulled up before the module is powered on normally. It is recommended to reserve test points.
50	GPIO1	DIO	General GPIO	VO_LDOIO	Internally connected in parallel with pin 63, cannot be used simultaneously.
51	GPIO3	DIO	General GPIO	VO_LDOIO	Internally

					connected in parallel with pin 64, cannot be used simultaneously.
52	GPIO4	DIO	General GPIO	VO_LDOIO	Internally connected in parallel with pin 28, cannot be used simultaneously. Internal GNSS connected, external use prohibited;
53	GPIO5	DIO	General GPIO	VO_LDOIO	Internal connection with pin 29 in parallel, cannot be used simultaneously. The internal GNSS is connected, external use is prohibited.
97,107	AGPIO5	DIO	General GPIO, can be maintained in sleep state.	LDO_AONIO	If not used, leave floating
99	AGPIO6	DIO	General GPIO, can be maintained in sleep state.	LDO_AONIO	Currently used as the internal pull-up power source for pin 62, external use is prohibited.
30,101	AGPIO3	DIO	General GPIO, can be	LDO_AONIO	If not used, leave

			maintained in sleep state.		floating
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12. Reserve interface

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
3~4,8,26,30~33,44,49,54~56,68,69,74~78,80~81,83~87,98,100,102~106,108~109	REV	-	-	-	Undefined pins are recommended to be left floating and not suggested for customer use (among them, 68, 69, 75, 86, 87, and 105 are prohibited from use and should be left floating).

Note: "*" indicates that the feature is under development.

Remarks

1.The AGPIO drive capability is limited, and the high-level output is prone to fluctuations. It is only suitable for signal level recognition and cannot be used as a peripheral power supply.

2.PIN8, PIN68, PIN69, PIN75, PIN86, PIN87, and PIN105 pins are recommended to be left floating and must not be grounded.

3.The driving capability of AGPIOWU is less than 30 μ A and can only be used as input, not as output.

4.PIN50 and PIN51 pins are not recommended for use; it is recommended to leave them floating.

2.5 Evaluation kit

Lierda can provide complete evaluation and development kits, including module adapter boards, Cat.1 & NB-IoT module universal development boards, etc. Feel free to contact us for inquiries.

3 Job characteristics

3.1 Working mode

The NT26-KCN E series module has multiple operating modes. The table below briefly describes several common operating modes of the module:

Table 3-1 Work Modes

Mode	Status Description
Normal working mode	Data Transmission: Network connection is normal. The module's power consumption depends on the network environment and data transmission rate.
	Idle state: The software is running normally. The module resides in the network but has no data interaction with the network.
Minimal functional mode	AT+CFUN=0 can set the module to minimum functionality mode, with both RF and USIM card not functioning.
Sleep mode	The module's power consumption will be reduced to extremely low levels, but the module can still receive paging, short messages, and TCP/UDP data.
Shutdown mode	VBAT power supply is not disconnected, but the software stops working.

Remarks

1. AT+CFUN related commands, please refer to the "AT Command Manual".
2. Detailed work modes and corresponding power consumption data refer to the PMU application documentation.

3.2 Sleep mode

The module can enter sleep mode through AT commands. In sleep mode, the module can reduce power consumption to an extremely low level, and the module's RF can still receive network information.

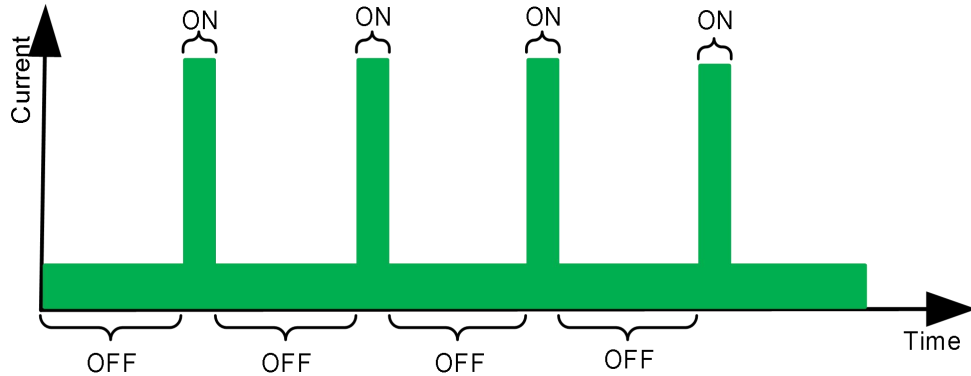


Figure 3.1 Power Consumption Status of the Module in Sleep Mode

In sleep mode, the module peripherals are powered down to reduce power consumption, but the RF remains in an intermittent receiving state and can still receive network information. When the module has a URC to report, it will wake up the host through the MAIN_RI pin.

3.2.1 Main serial port application scenarios

If the module and the host communicate via UART, the following two conditions must be met simultaneously to put the module into sleep mode:

- ◆ Execute the AT+QSClk=1 or AT+LSCLK=1 command (refer to the "AT Command Manual" for specific commands) to enable the sleep function;
- ◆ Ensure that MAIN_DTR remains at a high level or floating.

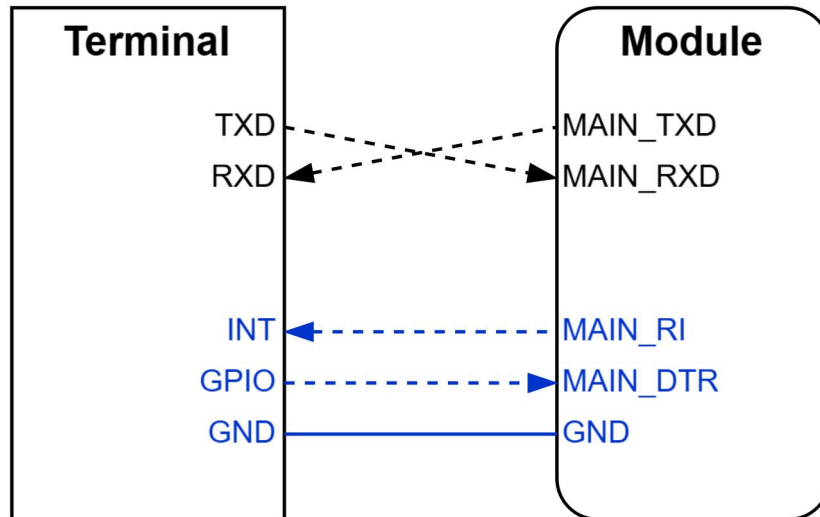


Figure 3.2 Application Circuit of Sleep Mode UART

- ◆ When the module has a URC to report, it notifies the host through MAIN_RI.
- ◆ The host pulling down MAIN_DTR can wake up the module.
- ◆ If the host continuously pulls down MAIN_DTR, it can prevent the module from entering sleep mode.
- ◆ If the sleep function is enabled using AT+QSCLK=2 or AT+LSCLK=2 (please refer to the "AT Command Manual" for specific commands), the module can be awakened by sending any command to the main UART interface.

3.2.2 USB application scenarios*

In USB application scenarios, the module needs to meet the following three conditions simultaneously to enter sleep mode:

- ◆ Execute AT+QSCLK=1/AT+LSCLK=1 (choose according to the "AT Command Manual");
- ◆ Ensure MAIN_DTR remains at a high level or floating;
- ◆ The host USB bus connected to the module's USB interface enters a suspended state.

3.2.2.1 Supports USB remote wake-up function*

The host supports USB suspend and wake-up as well as USB remote wake-up functions.

The connection between the module and the host, please refer to the diagram below:

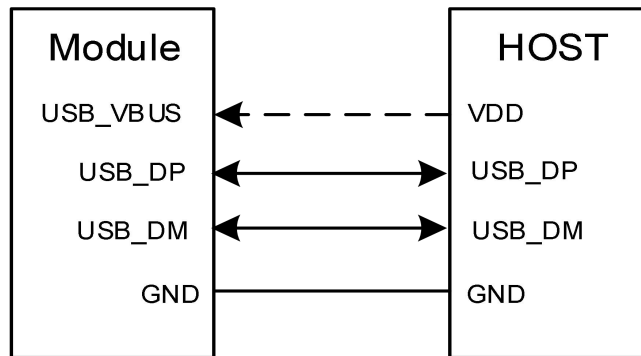


Figure 3.3 Sleep Application with USB Remote Wake-Up Function

- ◆ Sending data to the module via USB will wake up the module.
- ◆ When the module has a URC report, the module will send a remote wake-up signal through the USB bus to wake up the host.

3.2.2.2 Supports USB suspend and wake-up and MAIN_RI function*

If the host supports USB suspend and wake-up but does not support USB remote wake-up functionality, the host needs to be awakened by the MAIN_RI signal of the module.

The connection between the module and the host refers to the diagram below:

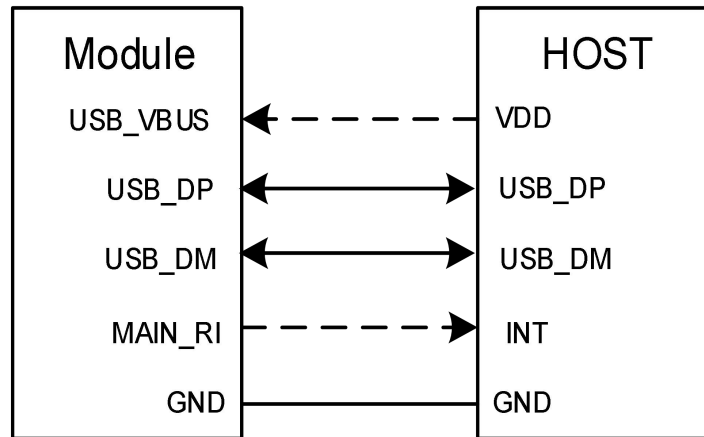


Figure 3.4 Sleep Application with MAIN_RI Function

- ◆ Sending data to the module via USB will wake up the module.
- ◆ When the module has a URC report, it can wake up the host through MAIN_RI.

3.2.2.3 Does not support USB suspend function*

When the host does not support the USB suspend function, the following three conditions must be met simultaneously to enable the module to enter sleep mode:

- ◆ Execute AT+QSCLK=1 or AT+LSCLK=1 (for specific commands, please refer to the "AT Command Manual");
- ◆ Ensure MAIN_DTR remains at a high level or floating;
- ◆ Disconnect USB_VBUS power supply.

If the host does not support the USB suspend function, the module can be put into sleep mode by disconnecting USB_VBUS through an external control circuit.

The connection between the module and the host refers to the diagram below:

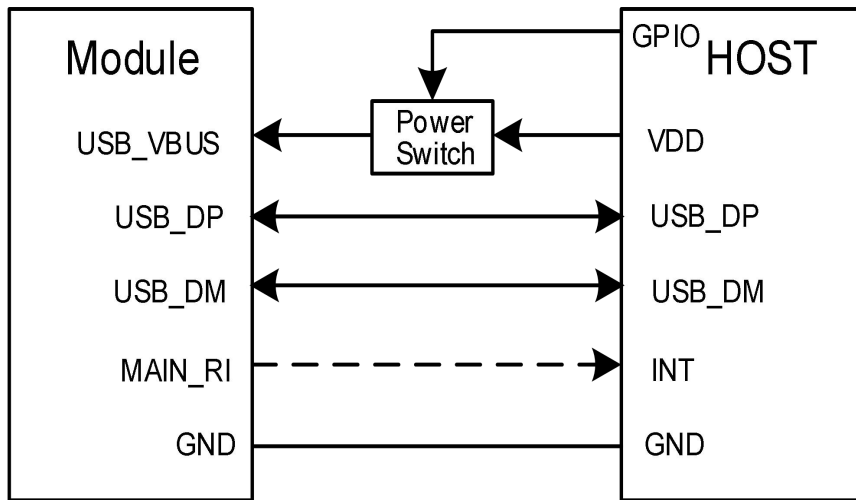


Figure 3.5 Sleep application without USB suspend function support

Remarks

Restoring USB_VBUS power supply can wake up the module.

3.3 Flight mode

When the module enters flight mode, the RF function cannot be used, and all RF-related AT commands are inaccessible. The module can be put into flight mode through the following methods:

You can set it by sending the AT+CFUN=<fun> command. The <fun> parameter can be set to 0 or 1.

- ◆ AT+CFUN=0: Minimum functionality mode (disable RF function and USIM card).
- ◆ AT+CFUN=1: Full functionality mode (default).

3.4 Power Design

3.4.1 Power Supply Introduction

The NT26-KCN E series module provides two VBAT pins for connecting an external power supply. Additionally, VDD_EXT is the module's external output power, which can be used as a reference level. The interface description is as follows:

Table 3-2 Power Pin Definitions

Pin number	Pin name	Description	Minimum value	Typical value	Maximum value	Unit
42,43	VBAT	Module power supply	3.3	3.8	4.5	V
24	VDD_EXT	Output power supply	-	1.8	-	V

The VBAT power supply range of the module is 3.3–4.5V, and it is necessary to ensure that the minimum input voltage does not drop below 3.3V due to voltage drops caused during module operation. The diagram below illustrates the VBAT voltage drop situation during burst transmission of the module:

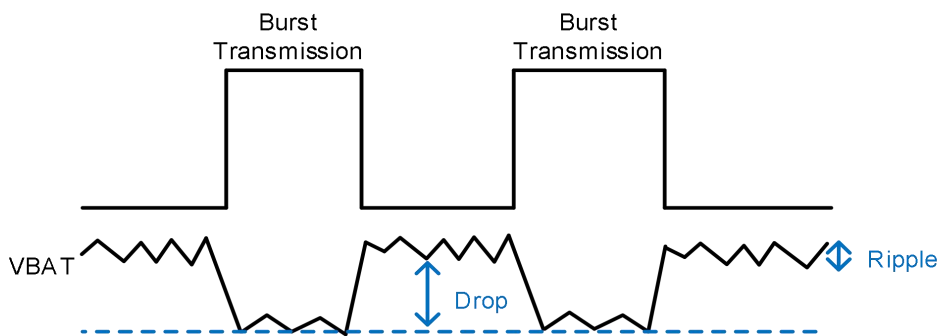


Figure 3.6 Burst Transmission Power Requirements

Remarks

In sleep mode, VDD_EXT will lose power.

3.4.2 Power Design

When the module operates at the maximum transmission power of the 4G network, the transient operating current in the current network can reach 1.2A, which may cause a voltage drop in the power supply. Under any circumstances, it is necessary to ensure that the module's power supply voltage remains above 3.3V; otherwise, unexpected situations such as module reboot may occur. To reduce voltage drops, it is recommended to use a 100uF filter capacitor with low ESR ($ESR < 0.7\Omega$). Additionally, it is suggested to reserve 3 multilayer ceramic capacitors (MLCC) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Furthermore, it is recommended to add a high-power TVS diode with low clamping voltage and high reverse pulse current I_{pp} at the VBAT terminal to enhance the module's surge voltage tolerance.

The power supply reference circuit is as follows:

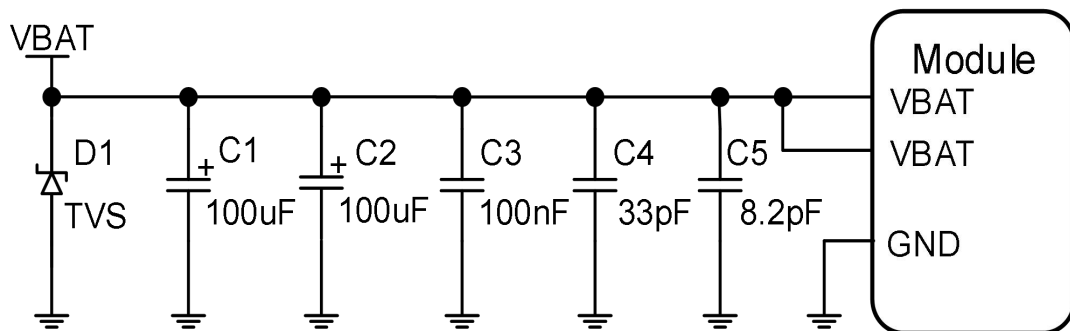


Figure 3.7 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider the trace width should be. It is recommended that the trace width be no less than 2mm. The GND plane in the power supply section should be as complete as possible, with more ground vias added. Additionally, the decoupling capacitor should be placed as close as possible to the VBAT pin of the module. Among them:

- ◆ D1 is a high-power low-clamping voltage TVS diode;
- ◆ C1 and C2 are 100uF low ESR tantalum capacitors, enhancing the power supply's continuous current capability and stabilizing the voltage.

◆ C3~C5 are filter capacitors with ceramic packages of 100nF, 33pF, and 8.2pF, respectively, used to eliminate high-frequency interference.

3.5 Boot up

3.5.1 Boot introduction

The module can be powered on through the PWRKEY pin, with the pin description as follows:

Table 3-3 PWRKEY Pin Description

Pin number	Pin name	I/O	Description	Remarks
7	PWRKEY	DI	The module is powered on, active low.	The default is high level.

The control logic of PWRKEY is as follows:

◆ In the shutdown state of the module, it can be powered on by pulling PWRKEY low for at least 1 second.

3.5.2 Power-on circuit reference design

It is recommended to use an open-collector drive circuit or directly control power-on through a button. To prevent static discharge caused by contact, a TVS should be placed near the button for ESD protection. The reference circuit is as follows:

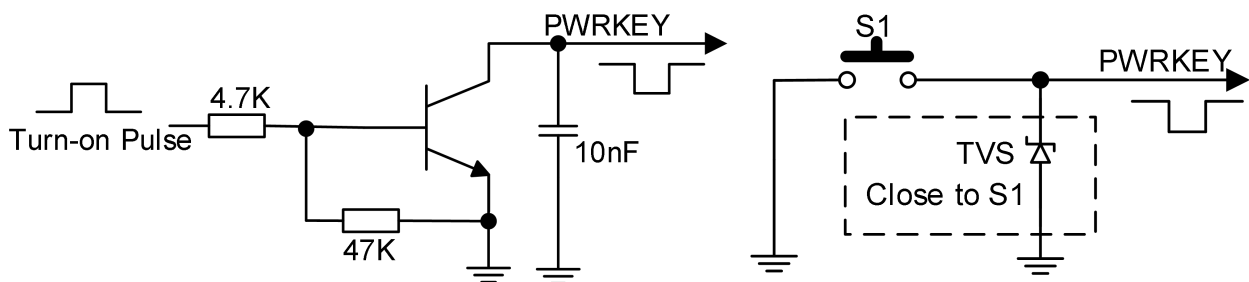


Figure 3.8 Power On/Off Circuit

3.5.3 Power-on sequence

The power-on sequence of the module is shown in the figure below:

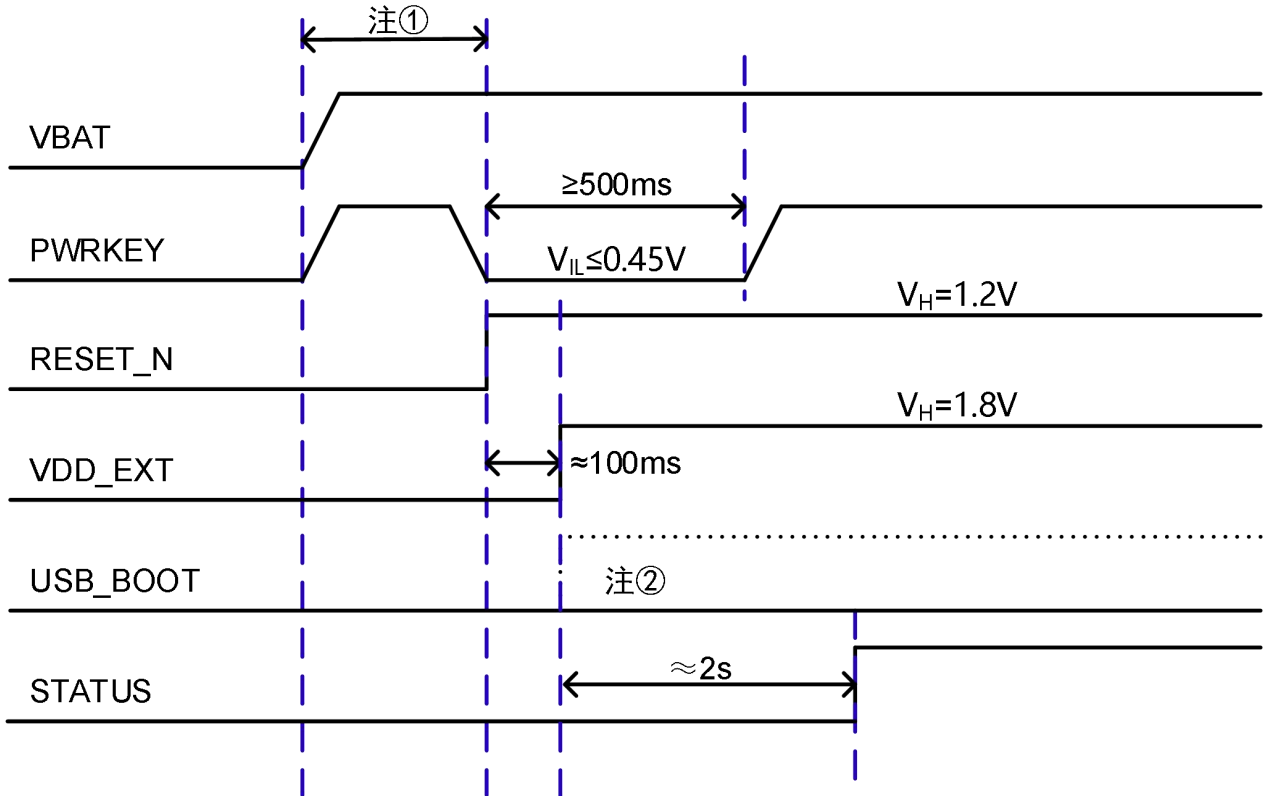


Figure 3.9 Power-On Startup Timing Diagram

Precautions

◆ The initial state of VBAT upon power-on needs to be less than 0.1V, and the time for VBAT to rise from 0V to 2.5V should be less than 10ms. Before pulling the PWRKEY pin low, ensure that the VBAT voltage is stable. It is recommended that the time interval between VBAT power-on and pulling the PWRKEY pin low be no less than 30ms.

◆ The USB_BOOT pin is pulled to VDD_EXT before the module is powered on, and the module will enter emergency download mode when powered on.

◆ If automatic power-on upon powering up is required and the shutdown function is not needed, PWRKEY can be directly pulled down to ground. It is recommended to use a 4.7k Ω pull-down resistor, or control PWRKEY to low level using GPIO before powering up the module and keep it low continuously.

◆ The module's main serial port sending the "Lierda" or "RDY" character indicates that the startup is complete, subject to the "AT Command Manual" for specifics.

3.6 Shutdown

The module can be properly shut down in the following ways:

- ◆ Control the module shutdown through the PWRKEY pin.
- ◆ Send the AT+QPOWD or AT+LPOWD command (please refer to the "AT Command Manual" for specific commands) to shut down.

3.6.1 PWRKEY shutdown

In the power-on state, pulling down PWRKEY for 650ms and then releasing it, the module executes the shutdown process, with the timing as follows:

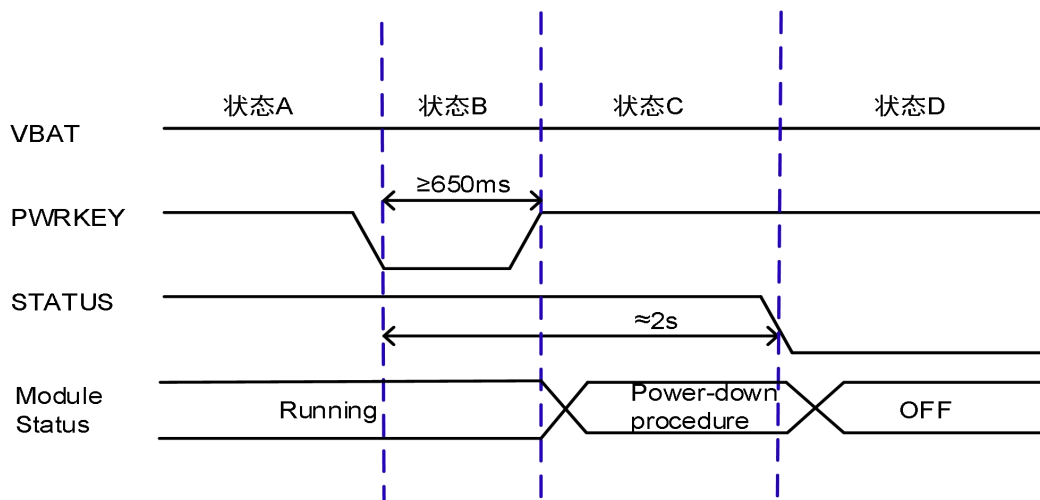


Figure 3.10 PWRKEY Shutdown Timing Diagram

PWRKEY shutdown sequence description:

- ◆ State A: Module working state;
- ◆ State B: The state of pulling down PWRKEY, with a duration of at least 650ms or more;
- ◆ State C: The module performs a system shutdown internally, taking approximately 2 seconds from pulling PWRKEY low to shutdown completion.
- ◆ State D: The module has entered shutdown mode.

3.6.2 AT command shutdown

Executing AT+QPOWD or AT+LPOWD (refer to the "AT Command Manual" for specific commands) can shut down the module. This operation has the same timing and effect as shutting down by pulling the PWRKEY low.

Precautions

◆ When the module is operating normally, do not immediately cut off the module's power supply, as this may damage the Flash data inside the module. It is recommended to first shut down the module via PWRKEY or AT command before disconnecting the power.

◆ When using the AT command to shut down, please ensure that PWRKEY remains at a high level after the shutdown command is executed; otherwise, the device may restart after shutting down. Disconnecting the power at this time poses a risk of damaging the Flash.

◆ If using the method of keeping PWRKEY grounded to power on, it is recommended to execute the shutdown process using AT+QPOWD/AT+LPOWD (choose according to the "AT Command Manual") before disconnecting the power supply.

3.7 Reset

3.7.1 Reset Introduction

The RESET_N pin can be used for module reset. Pulling the RESET_N pin low for at least 300ms and then releasing it can reset the module. The RESET_N pin is described as follows:

Table 3-4 Reset Pin Description

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
15	RESET_N	DI	Module reset	VILmax=0.5V	Internal weak pull-up, active low, leave floating if not used.

3.7.2 Reset circuit reference design

The RESET_N signal is relatively sensitive to interference, so it is recommended that the module interface routing should be as short as possible and should be ground-shielded. The typical hardware circuit reference application is as follows, and the two circuit diagrams can be selected based on actual needs:

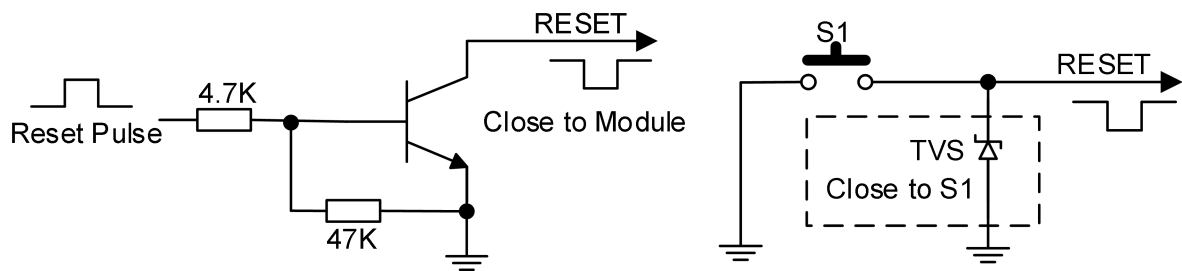


Figure 3.11 Reset Circuit Reference Design

3.7.3 Reset timing

The reset timing of the module is as follows:

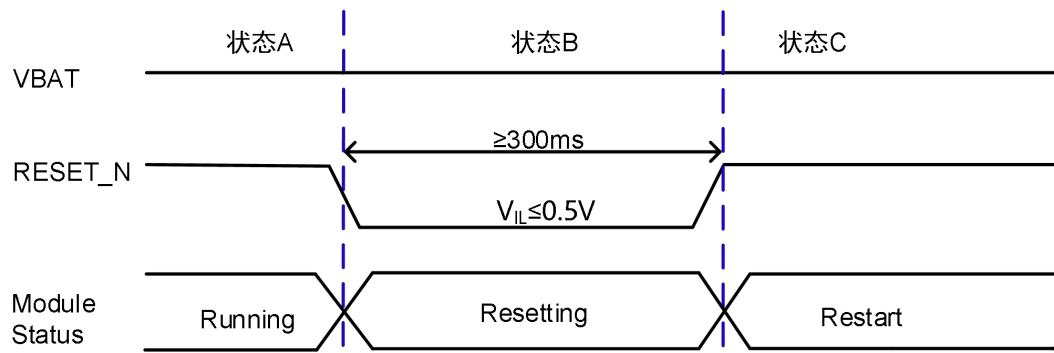


Figure 3.12 Reset Timing Diagram

Reset timing description:

- ◆ State A: Module normal working state;
- ◆ State B: The state of pulling RESET_N low, with a duration of at least 300ms or more;
- ◆ Status C: The module begins to restart.

4 Application Interface

The NT26-KCN E series module provides a wealth of peripheral functions, mainly including the following application interfaces:

- ◆ 1*USB2.0 interface*
- ◆ 3*UART interface
- ◆ 2*USIM interfaces
- ◆ 1*I2C Interface
- ◆ 1*Analog Audio Interface

4.1 USB interface

The NT26-KCN E series module supports a USB 2.0 interface, complies with the USB 2.0 specification, and supports full-speed (12 Mbps) and high-speed (480 Mbps) modes. The module only supports USB slave mode. This interface can be used for AT command sending, data transmission, software debugging, and firmware upgrades. The interface definition is as follows:

Table 4-1 USB Interface Pin Description

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
59	USB_DP	DIO	USB differential data (+)	-	Complies with USB 2.0 specifications. Requires 90 Ω differential impedance. Test points must be reserved. Leave them floating if not used.
60	USB_DM	DIO	USB differential data (-)	-	Complies with USB

					2.0 specifications. Requires 90 Ω differential impedance. Test points must be reserved. Leave them floating if not used.
61	USB_VBUS	DI	USB Insert Wake-up Module	Vmax = 5.25 V Vmin = 3.0 V	In sleep mode, USB insertion will wake up the module.
82	USB_BOOT	DI	USB mode configuration interface	VO_LDOIO	Internal weak pull-down, active high. Do not pull up this pin before the module powers on normally. It is recommended to reserve a test point; leave it floating if not used.

4.1.1 USB Circuit Reference Design

The USB interface of the module can be connected to a USB connector or the USB interface of an MCU. The reference schematic for connecting to the USB connector is as follows:

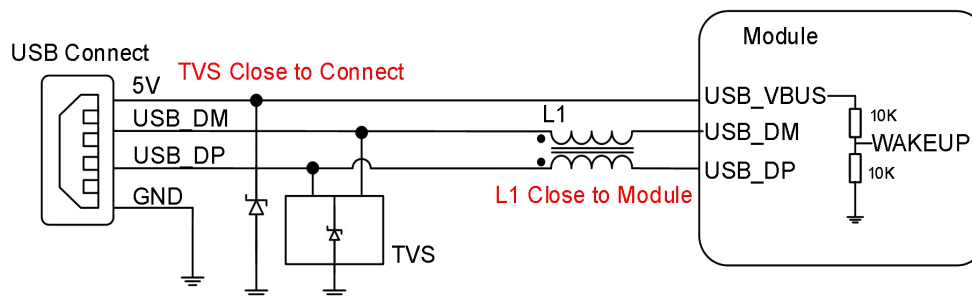


Figure 4.1 USB Interface Reference Design

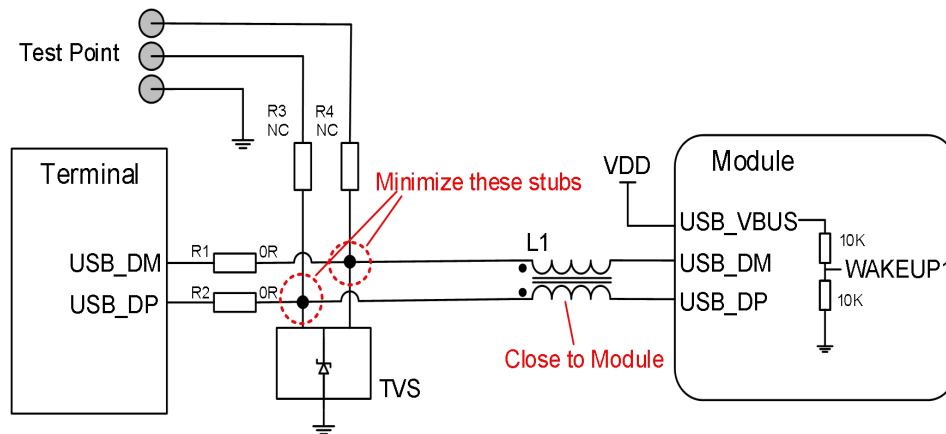


Figure 4.2 USB Connection MCU Reference Design

Precautions

◆ To ensure USB performance, the USB_DP and USB_DM traces should maintain a differential impedance of 90Ω , and the surrounding area needs to be shielded with ground. It is recommended to connect a common-mode inductor L1 between the MCU and the module to prevent EMI interference from the USB signal. L1 should be placed close to the module.

◆ USB routing should stay away from crystals, oscillators, magnetic devices, RF signals, and strong signal areas. It is recommended to use inner-layer differential routing with ground shielding on all sides.

◆ If the module's USB interface and MCU are connected via a connector, a TVS diode should be added at the interface for protection. Special attention should be paid to the selection of ESD protection devices, ensuring that their parasitic capacitance does not exceed 2pF, and they should be placed as close to the USB interface as possible.

◆ Reserve resistors R1~R4. Normally, attach R1~R2 and do not attach R3~R4 to connect the MCU properly. During debugging, attach R3~R4 and do not attach R1~R2 to switch to the test points. To meet the signal integrity requirements of the USB data

lines, keep the branch lines connected to the test points as short as possible.

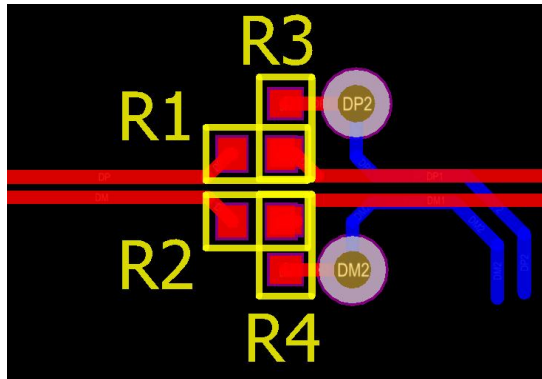


Figure 4.3 USB Jumper Routing Diagram

4.1.2 USB data transmission

When the module is powered on or reset and USB_BOOT is detected as low level, the USB port will enumerate two serial ports for AT command transmission and Log output.

4.1.3 USB firmware download

When the module is powered on or restarted by reset, if USB_BOOT is detected as high level, the USB port will enumerate a dedicated download serial port. At this time, the download tool can perform firmware upgrades through the USB interface. If the download tool is not connected within 15 seconds, the module will re-enumerate 2 serial ports according to the default USB configuration.

The reference design for the USB_BOOT interface is as follows:

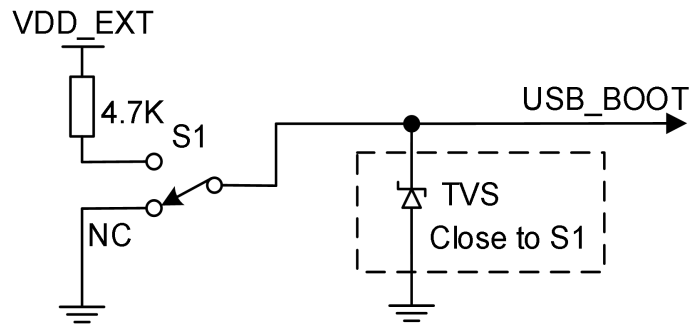


Figure 4.4 USB_BOOT Reference Design

4.2 UART Communication

The module provides 3 general-purpose asynchronous transceivers: the main serial port, the debug serial port, and the auxiliary serial port. The baud rate of the main serial port is configurable, and the debug serial port is only used for debugging and testing purposes. The serial port pin definitions are as follows:

Table 4-2 Serial Port Pin Definitions

Interface	Pin number	Pin name	Description	Remarks
Main serial port	17	MAIN_RXD	The main serial port receives data.	Non-sleep state: VDD_EXT power domain; Sleep state: Switch to internal AON power domain
	18	MAIN_TXD	The main serial port sends data.	If not used, leave floating
	19	MAIN_DTR	The main serial port data terminal is ready.	In sleep mode, it can serve as a wake-up pin.
	20	MAIN_RI	The main serial port outputs a ringing	If not used, leave floating

			prompt.	
	21	MAIN_DCD*	The main serial port outputs carrier detection.	Not supported for now
	22	MAIN_CTS*	The main serial port DTE clears sending.	Internally connected in parallel with pin 57, cannot be used simultaneously.
	23	MAIN_RTS*	The main serial port DTE requests to send.	Internally connected in parallel with pin 58, cannot be used simultaneously.
Debug serial port	38	DBG_RXD	Debugging serial port data reception	If not used, leave floating
	39	DBG_TXD	Debugging serial port data transmission	If not used, leave floating
Auxiliary serial port	28	AUX_RXD*	Assist serial port in receiving data.	Not supported for now.
	29	AUX_TXD*	Assist the serial port in sending data.	Not supported for now.

Precautions

◆ The serial port interface belongs to the VO_LDOIO power domain, with a default voltage level of 1.8V.

◆ The VO_LDOIO power domain will lose power in sleep mode, and MAIN_RXD will switch to the internal 1.2V power supply.

◆ The MAIN_RXD pin has an internal pull-up, so no external pull-up resistor is needed, and it has backflow prevention functionality.

◆ Pay attention to the issue of level consistency during serial port usage; otherwise, leakage current is likely to occur.

4.2.1 Serial Port Application

Main serial port features:

- ◆ Used for AT command communication and data transmission, with a default baud rate of 115200bps (8-N-1);

- ◆ Configurable as:
4800/9600/19200/38400/57600/115200/230400/460800/921600bps;

- ◆ Used for firmware upgrades, with a default baud rate of 921600bps during upgrades.

- ◆ Supports low-power wake-up module function.

Features of debugging serial port:

- ◆ Through the dedicated tools provided by the platform, log information can be viewed for software debugging, with a default of 3000000bps.

The schematic diagram of the serial connection method is as follows:

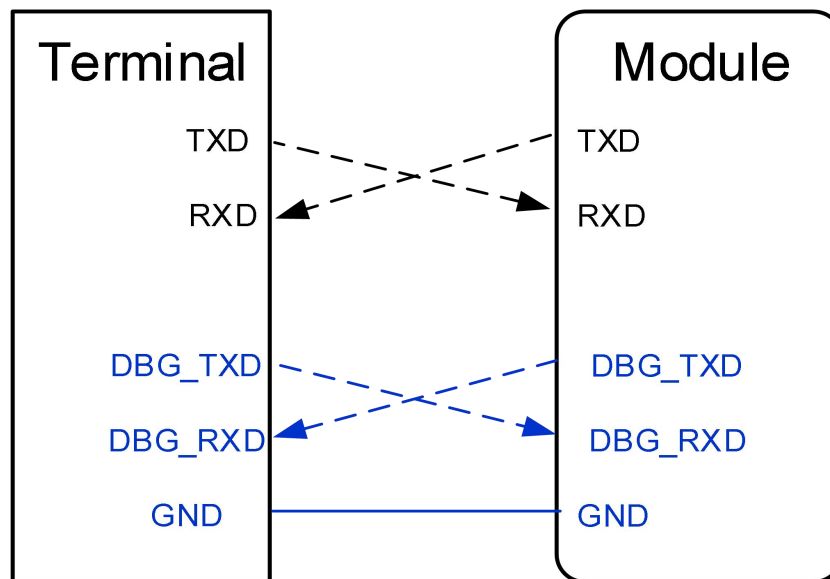


Figure 4.5 Serial Port Connection Diagram

4.2.2 Serial Port Circuit Reference Design

The main factors to consider for a suitable serial port level conversion circuit include: whether it meets the working rate of the serial port, and in scenarios with low power

consumption requirements, whether its power consumption meets the requirements, etc. It is recommended to refer to the following level matching circuit:

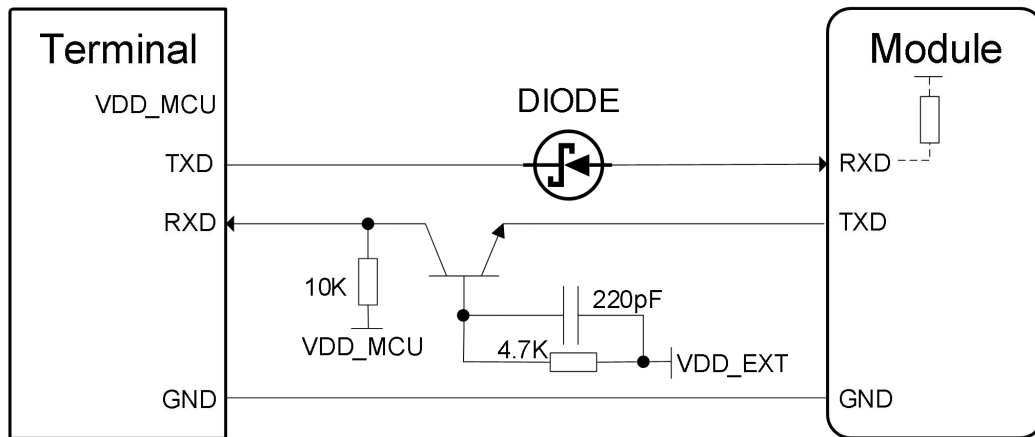


Figure 4.6 Transistor Level Conversion Reference Circuit

Recommend transistors for reference:

Brand: CJ Model: S8050 J3Y Package: SOT-23

Recommend diodes for reference (need to note that the forward voltage of the diode $\leq$ 0.3V):

Brand: LRC

Model: LRB520S-30T1G

Package: SOD-523

4.3 USIM card interface

The module includes 2 external USIM card interfaces. Card 1 complies with ETSI and IMT-2000 standards, supporting 1.8 V and 3.0 V USIM cards, while Card 2 supports 1.8 V USIM cards by default. The pin description is as follows:

Table 4-3 External USIM Card Interface Pin Definition

Pin number	Pin name	Description	DC characteristics
11	USIM_DATA	SIM card data cable	VO_LDOSIM

12	USIM_RST	SIM card reset line	VO_LDOSIM
13	USIM_CLK	SIM card clock line	VO_LDOSIM
15	USIM_VDD	Power supply for SIM card	Vnorm=1.8/3.0V
79	USIM_DET	SIM card status signal detection	Hot-swappable detection interface for SIM card slot
62	USIM2_CLK	SIM2 (secondary card) clock line	VO_LDOIO
63	USIM2_RST	SIM2 (secondary card) reset line	Internal and pin 50 are connected in parallel and cannot be used simultaneously.
64	USIM2_DATA	SIM2 (secondary card) data cable	Internally connected in parallel with pin 51, cannot be used simultaneously.
65	USIM2_VDD	Power supply for SIM2 (secondary card).	The same power supply as USIM_VDD.

The module supports USIM card hot-plugging through USIM1_DET and supports both high/low level detection. This function is disabled by default and can be configured via AT+QSIMDET. If the USIM card detection function is not needed, USIM_DET can be left floating.

4.3.1 USIM card circuit reference design

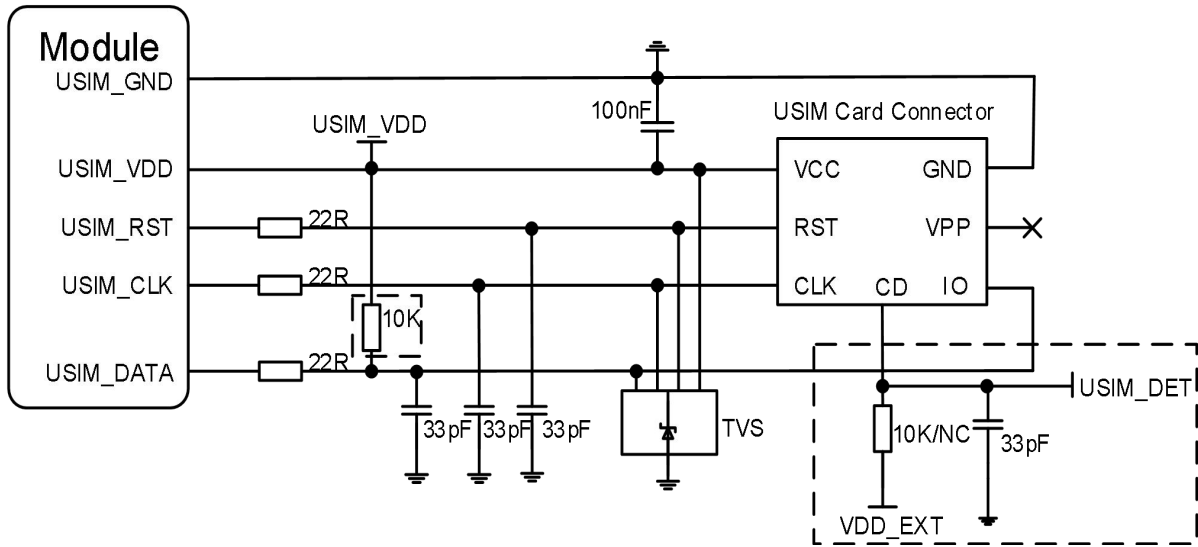


Figure 4.7 6PIN SIM Card Reference Circuit

Precautions

To ensure that the power consumption in sleep mode meets the design standards, it is recommended to use a SIM card holder with a normally closed CD pin.

The hardware design circuit of USIM2 can also refer to the above design, specifically referring to the "Lierda NT26-KCN E Series Hardware Reference Design Manual".

4.3.2 USIM card circuit design considerations

To ensure the reliability and availability of the SIM card in applications, please follow the following USIM circuit design guidelines:

- ◆ When laying out, try to place the SIM card as close to the module as possible, and keep the signal line length preferably within 200mm.
- ◆ The SIM card signal line should be kept away from RF routing and VBAT power lines.
- ◆ The GND wiring of the SIM card should be short and thick, ensuring the wiring

width is not less than 0.5mm;

◆ The decoupling capacitor of USIM_VDD should not exceed 1uF, and the capacitor should be placed close to the VCC of the external SIM card.

◆ To avoid signal crosstalk between USIM_DATA and USIM_CLK, their routing should not be too close, and ground shielding needs to be added between the two traces. Additionally, to mitigate the impact of excessive trace length, USIM_DATA generally requires a pull-up resistor to USIM_VDD to enhance driving capability. If the trace is too long, it is recommended to reserve an external 10K resistor near the card slot. Furthermore, the USIM_RST signal also needs ground shielding protection.

◆ To ensure good ESD protection performance, it is recommended to add a TVS diode to the external USIM card pins, with the selected TVS diode's parasitic capacitance not exceeding 15pF. A 22-ohm resistor should be connected in series on the signal line between the module and the SIM card to suppress stray EMI and enhance ESD protection. Additionally, a parallel 33pF capacitor is used to filter out RF interference. The related resistors, capacitors, and TVS should be placed close to the USIM card slot.

4.4 I2C interface

The module provides 1 I2C interface, described as follows:

Table 4-4 I2C Interface Pin Definition

Pin number	Pin name	Description	DC characteristics	Remarks
66	I2C_SDA	I2C data	VO_LDOIO	I2C data signal requires an external 4.7K pull-up resistor, otherwise it will be left floating.
67	I2C_SCL	I2C Clock	VO_LDOIO	The I2C clock signal requires an external 4.7K pull-up resistor; otherwise, it will be left floating.

4.5 Analog audio interface

4.5.1 Introduction to Analog Audio

Table 4-5 SPK Pin Definition

Pin number	Pin name	Description	DC characteristics	Remarks
5	SPK_P	Audio positive differential output	VO_LDOIO	Internally connected in parallel with pin 82, cannot be used simultaneously.
6	SPK_N	Audio negative differential output	GND	If not used, leave floating

Remarks

The audio output channel can be used as a speaker (if the output power is insufficient, it is recommended to add an external audio amplifier device) or an earpiece output channel. To enhance anti-interference capability, it is recommended to use differential output.

4.5.2 Analog audio reference application

Through an external audio PA, high-power speaker playback can be achieved. A common application reference circuit is as follows:

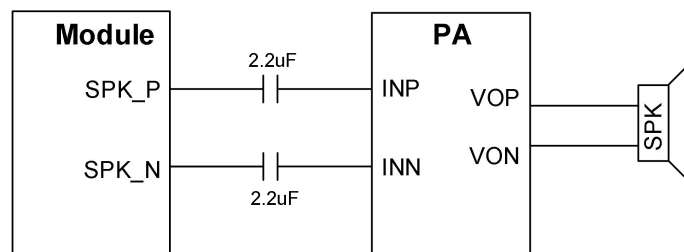


Figure 4.8 SPK Application Reference Circuit

In practical applications, the selected PA may vary, and the circuit will also differ

accordingly. The specific circuit can be found in the "Lierda NT26-KCN E Series Hardware Reference Design Manual" document.

4.5.3 Audio Application Notes

The severity of high-frequency interference during GSM transmission usually depends on the application design. In some cases, TDD noise under EGSM900 is more severe, while in other cases, TDD noise under DCS1800 is more severe. Therefore, the required filtering capacitors can be selected based on actual test results. The placement of RF filtering capacitors on the PCB should be as close as possible to the audio devices or audio interfaces, and the routing should be kept as short as possible, passing through the filtering capacitors first before reaching other points.

To reduce radiation interference, the antenna position should be kept as far away as possible from audio components and audio traces; power traces and audio traces should not run parallel, and power traces should be kept as far away as possible from audio traces.

4.6 ADC interface

The module provides 2-channel 12-bit analog-to-digital conversion input interfaces (ADC interfaces). When routing the ADC interfaces, it is recommended to apply ground shielding to improve the voltage measurement accuracy of the interface. The pin definitions are as follows:

Table 4-6 ADC Pin Definitions

Pin number	Pin name	Description	DC characteristics	Remarks
9	ADC0	General ADC Interface	Input voltage range: 0~1.05V	If not used, leave floating
96	ADC1	General ADC Interface		If not used, leave floating

Using AT+QADC=<port> can read the voltage value of each pin on the ADC interface:

◆ AT+QADC=0 or AT+LADC=0 (please refer to the "AT Command Manual" for specific

commands): can be used to read the voltage value of ADC0.

- ◆ AT+QADC=1 or AT+LADC=1 (please refer to the "AT Command Manual" for specific commands): Can be used to read the voltage value of ADC1.

Table 4-7 ADC Characteristics

Parameter	Minimum value	Typical value	Maximum value	Remarks
ADC voltage range	0	-	1.05	V
ADC input impedance	0.26	-	0.75	Ω
ADC precision	-	12	-	bits

Precautions

◆ If the collected voltage is greater than 1.05V, it is recommended to use a resistor voltage divider circuit for the ADC pin input. The resistance value of the voltage divider should not exceed 100K Ω , and it is recommended to use resistors with an accuracy of 1% or higher; otherwise, the measurement accuracy of the ADC will be reduced.

◆ Reserve a 1nF capacitor across the voltage-dividing resistors for grounding during design, which can be left unmounted by default.

4.7 Status indication signal

The module has one network status indicator and one operation status indicator interface, defined as follows:

Table 4-8 Status Indicator Pin Definitions

Pin numbe	Pin name	Description	DC characteristics	Remarks
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r				
16	NET_STATUS	Network status indication	VDD18AON	If not used, leave floating
25	STATUS	Operating status indication	LDO_AONIO/ VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, will not lose power during sleep.

4.7.1 Network status indication

The NET_STATUS pin is used to indicate the network status of the module, and the logic levels under different states are as follows:

Table 4-9 Status Indicator Pin Definitions

Pin name	Logic level	Network status
NET_STATUS	Slow flash (200ms on / 1800ms off)	Network search status
	Slow flash (1800ms high/200ms low)	Standby state
	Flash (125ms High/125ms Low)	Data Transmission Mode

4.7.2 Operating status indication

The STATUS pin is used to indicate the operating status of the module. When the module is powered on normally, STATUS will output a high level, and it remains effective even in sleep mode.

4.7.3 Indicator light circuit reference design

Both the NET_STATUS and STATUS pins can be used to drive LED indicators, with the reference design as follows:

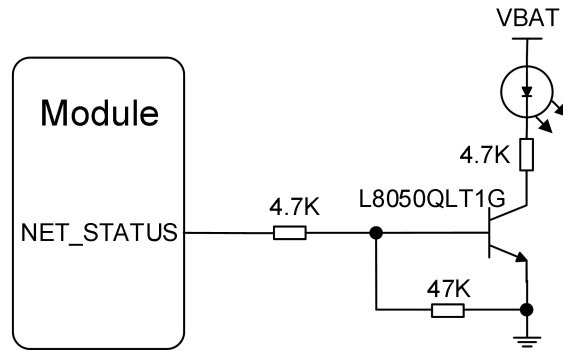


Figure 4-9 Network Status Indicator Reference Circuit

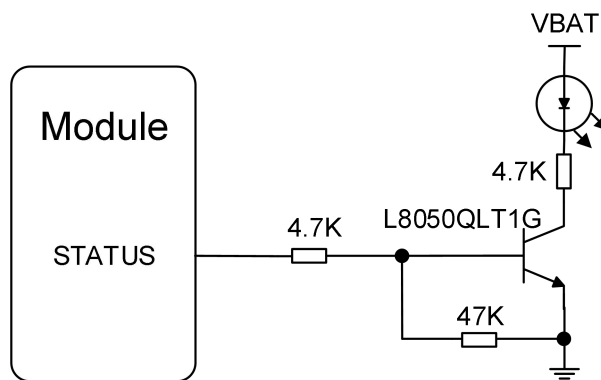


Figure 4-10 Module Status Indicator Light Reference Circuit

5 RF characteristics

5.1 Introduction to RF Function

The NT26-KCN E series module provides one RF interface with a characteristic impedance of 50Ω . The interface description is as follows:

Table 5-1 RF Interface Pin Definition

Pin number	Pin name	I/O	Description	Remarks
35	ANT_MAIN	AIO	Main antenna interface	50Ω characteristic impedance

5.2 Cellular network

5.2.1 Operating frequency band

The working frequency bands supported by the NT26-KCN E series module are as follows:

Table 5-2 Working Frequency Bands

Operating frequency band	Transmission (MHz)	Reception (MHz)
LTE-FDD B1	1920~1980	2110~2170
LTE-FDD B3	1710~1785	1805~1880
LTE-FDD B5	824~849	869~894
LTE-FDD B8	880~915	925~960
LTE-TDD B34	2010~2025	2010~2025
LTE-TDD B38	2570~2620	2570~2620
LTE-TDD B39	1880~1920	1880~1920
LTE-TDD B40	2300~2400	2300~2400

LTE-TDD B41	2535~2675	2535~2675
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Precautions

LTE-TDD B41 only supports 140M (2535-2675MHz).

5.2.2 Transmission power

The transmit power of the NT26-KCN E series module is as follows:

Table 5-3 Conducted Power

Frequency band	Maximum value	Minimum value	Remarks
LTE-FDD B1 (10 MHz)	23dBm±2.7dB	<-40dBm	Compliant with Cat.1 protocol in 3GPP Rel-13 and Rel-14.
LTE-FDD B3 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-FDD B5 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-FDD B8 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B34 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B38 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B39(10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B40 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B41 (10 MHz)	23dBm±2.7dB	<-40dBm	

5.2.3 Reception sensitivity

The reception sensitivity of the NT26-KCN E series module is as follows:

Table 5-4 Conduction Reception Sensitivity Under Single Transmission (Throughput ≥95%)

Frequency band	Receiving sensitivity (10M)	3GPP Standard (10M)
LTE-FDD B1 (10 MHz)	-96.6dBm	-96.3dBm
LTE-FDD B3 (10 MHz)	-98.3dBm	-93.3dBm
LTE-FDD B5 (10 MHz)	-98.3dBm	-94.3dBm

LTE-FDD B8 (10 MHz)	-99.5dBm	-93.3dBm
LTE-TDD B34 (10 MHz)	-100.4dBm	-96.3dBm
LTE-TDD B38 (10 MHz)	-100dBm	-96.3dBm
LTE-TDD B39(10 MHz)	-100dBm	-96.3dBm
LTE-TDD B40 (10 MHz)	-100.9dBm	-96.3dBm
LTE-TDD B41 (10 MHz)	-99.7dBm	-94.3dBm

5.3 Antenna Reference Circuit Design

When users use the NT26-KCN E series module, a π -type matching circuit needs to be added between the module's RF antenna interface and the user's baseboard antenna interface. The typical antenna matching circuit and initial parameters are shown in the figure below:

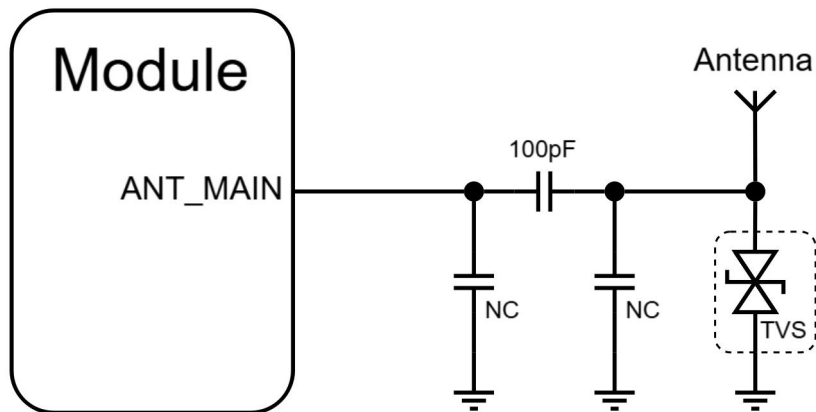


Figure 5.1 RF Antenna Reference Circuit

Precautions

- ◆ The crosstalk adopts 100pF, the capacitor position is not installed by default, and the device package is recommended to use 0201 or 0402 package.
- ◆ If an external antenna is used, or if the user can touch the antenna, it is

recommended to reserve a TVS diode to enhance electrostatic protection. Since the parasitic capacitance of the TVS diode may affect antenna performance, a model with parasitic capacitance $\leq 0.5\text{pF}$ should be selected. After adding the TVS diode, it is recommended to retune the antenna.

5.4 Guidelines for RF Signal Line Routing

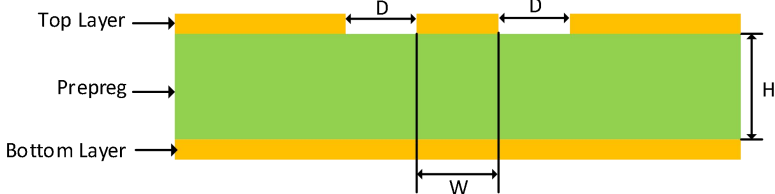
The routing between the module antenna interface and the user antenna must meet the 50-ohm characteristic impedance requirement, and the RF routing should be as short as possible to ensure minimal insertion loss in the RF routing.

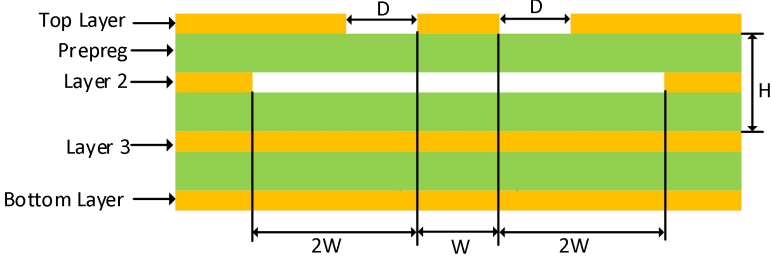
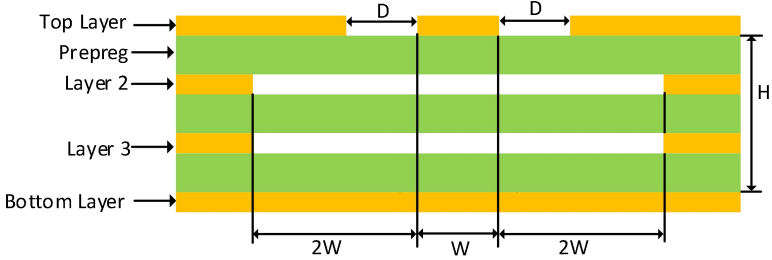
Under normal circumstances, the impedance of the RF signal line is determined by the dielectric constant ϵ_r of the material, the trace width W , the spacing to ground D , and the thickness of the reference ground plane H .

In the field of IoT applications, the design of PCB characteristic impedance is usually implemented using a coplanar waveguide method, which helps provide better shielding for RF signal lines and achieves a higher level of integration for compact design.

Common PCB coplanar waveguide designs include the following:

Table 5-5 Common PCB Coplanar Waveguide Structure Design Description

Serial Number	Pin Name	Remarks
1	 The diagram illustrates a 2-layer PCB coplanar waveguide structure. It shows a cross-section with three layers: Top Layer (yellow), Prepreg (green), and Bottom Layer (yellow). The top layer has two signal traces separated by a gap of width D. The bottom layer is a solid ground plane. The thickness of the prepreg layer is H. The width of the signal traces is W.	2-layer board The reference site is the L2 layer.

2		4-layer board The reference location is the L3 layer. L2 layer hollowing
3		4-layer board The reference site is the L4 layer. L2, L3 layer hollowing

During the Layout process, a 50-ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on the actual situation, choose the appropriate line width, line spacing, and stack-up to achieve the best design results.

Take the calculation of a coplanar waveguide with a 2-layer board and 1.6mm board thickness as an example:

◆ Choose the appropriate PCB material. Once the material is selected, the dielectric constant ϵ_r is basically determined (4.6 in this case).

◆ Choose an appropriate line width. A relatively good method is to select a line width that matches the size of the device pad (note that it should not be too thin; in this example, 0.65mm is chosen).

◆ According to the selected board thickness, line width, dielectric constant, etc., the ground spacing can be roughly calculated. If the line spacing is not suitable, the line width can be adjusted to modify the spacing (calculated as 0.14mm, basically appropriate).



Figure 5.2 Reference for the calculation method of 50-ohm impedance

In the actual Layout process, the following suggestions are provided for reference:

- ◆ In the π -type circuit, the three matching reserved components are placed close to the antenna, and the bypass components are recommended to be distributed on both sides of the RF line.
- ◆ Irregular vias should be placed on both sides of the RF trace's GND plane, and there must be a complete GND plane beneath the entire RF trace space.
- ◆ There should be no other routing below the RF line to avoid affecting RF performance or other circuits.

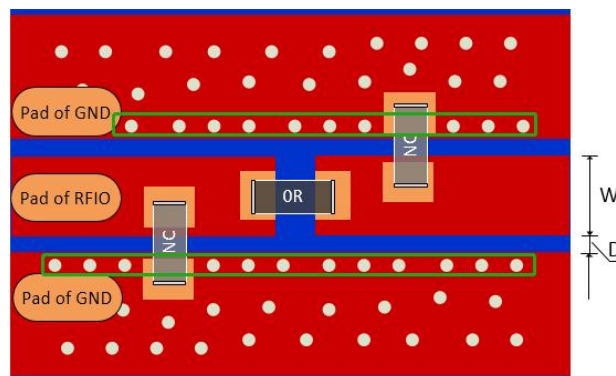


Figure 5-3 RF Trace Layout Design Schematic Diagram

The rationality of RF routing can be directly reflected in the conducted test of the module. To ensure that the complete product can achieve maximum performance, it also requires coordination with antenna design. To better meet the needs of antenna design, the following requirements are expected to be achieved in PCB design. The following provides

guidance for complete machine PCBs with different layer counts:

◆ When the product PCB is designed with 2 layers, it is best for both the TOP and BOTTOM LAYER directly below the module to be GND layers. The traces that need to be routed from the module should avoid passing directly below the module and should be routed from the outside of the module.

◆ When the product PCB is designed with 4 layers, it is recommended that the traces drawn out from the module be routed on the third or fourth layer, reserving the first and second layers as complete GND reference layers for the module.

5.5 Antenna design requirements

The RF performance of the module is also affected by the antenna, and the selection of the antenna needs to meet the following requirements:

- ◆ Select an antenna that matches the working frequency band of the module.
- ◆ Require the characteristic impedance of the antenna to be 50 ohms to reduce the loss at the connection between the RF line and the antenna.
- ◆ The smaller the insertion loss within the working frequency band, the better, such as $VSWR \leq 2$ and return loss $\geq 10\text{dB}$, etc.;

Common antennas suitable for CAT1 scenarios are shown in the figure below:



Figure 5.4 Common Antenna Types for CAT1

5.6 RF connector recommendation

In some application scenarios requiring customized antennas, such as environments where the device is shielded by an iron shell, an IPX or SMA connector can be reserved on the motherboard, and the antenna can be connected to the outside of the device via a radio frequency cable to achieve better performance.

Table 5.6 Commonly Used RF Connectors

Serial Number	Product Name	Picture	Remarks
1	IPX seat		Connect the RF cable.
2	SMA connector		Can connect to RF cable or antenna.

6 GNSS performance

Table 6-1 GNSS Antenna Interface Pin Definition

Pin number	Pin name	Description	Remarks
2	GNSS_ANT	GNSS main antenna interface	Supports GPS and BDS positioning systems. 50 Ω characteristic impedance; Supports active GNSS antenna; Leave it unused and keep it suspended.

Table 6-2 Working Frequency Bands

Type	Frequency band	Unit
GPS*	1575.42 ± 1.023	MHz
BDS	1561.098 ± 2.046	MHz

Precautions

The NT26K2E1-0G module only supports a single BeiDou frequency band and does not support the GPS frequency band.

6.1 GNSS Performance

Table 6-3 GNSS Performance

Parameter	Conditions	Typical value
Sensitivity (dBm)	Cold start	-149dBm
	Recapture	-159dBm
	Track	-164dBm

First positioning time	Cold start @ -130dBm	32s
	Warm start @ -130dBm	28.1s
	*Hot Start @ -130dBm	1s
Positioning accuracy (m) CEP95%	Simulator signal (static)	0.6m
CEP-50	@Pure Space Environment, Open Rooftop	1m
Power consumption	Capture	66mA
	Continuous tracking	26mA
	Backup power mode	14.72uA

Remarks

Tracking sensitivity: The lowest GNSS signal value at the corresponding antenna port when the module continuously positions for 3 minutes.

Reacquisition Sensitivity: The minimum GNSS signal value at the corresponding antenna port when the module repositions within 3 minutes after losing lock.

Cold start sensitivity: The minimum GNSS signal value at the corresponding antenna port when the module executes a cold start command and achieves positioning within 3 minutes.

6.2 Reference Design

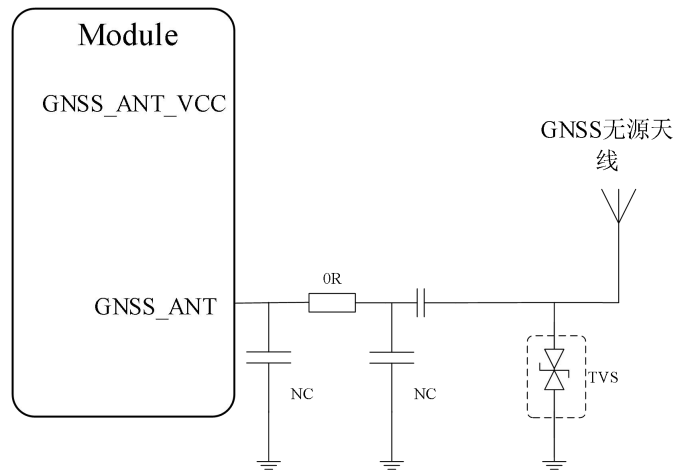


Figure 6-1 GNSS Passive Antenna Reference Circuit

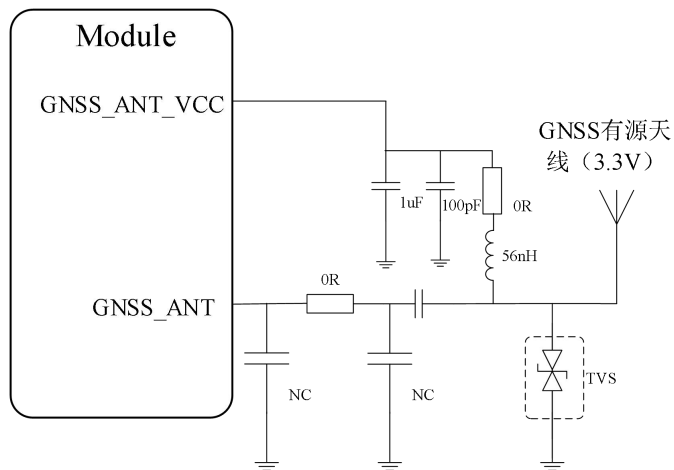


Figure 6-2 GNSS Active Antenna Reference Circuit

Remarks

If an active antenna is used, the module GNSS_ANT_VCC power supply can be used, with a VBAT power supply range of 3.3V.

If a passive antenna is used, the internal control GNSS antenna power can be turned off. According to the antenna form, especially antennas that are prone to introducing static electricity, in order to prevent static electricity from damaging the module, a TVS diode ($\leq 0.5\text{pF}$) or an inductor ($\geq 47\text{nH}$) can be reserved near the antenna port.

7 Electrical performance and reliability

7.1 Absolute Maximum Ratings

The absolute maximum ratings of the module interface are described as follows:

Table 6-1 Absolute Maximum Ratings

Parameter	Minimum value	Maximum value	Unit
VBAT	-0.3	5.0	V
USB_VBUS voltage	-0.3	5.25	V
Digital interface voltage	-0.3	3.6	V
VBAT current	-	1.2	A

Precautions

Permanent damage to the module may occur if the operating conditions exceed the absolute maximum ratings.

7.2 Power rating

The electrical characteristics of the module power interface are described as follows:

Table 6-2 Electrical Characteristics of Power Interface

Parameter	Description	Condition	Minimum value	Typical value	Maximum value	Unit
VBAT	Normal supply voltage of the module (1)	The actual input voltage must be within this range.	3.3	3.8	4.5	V
VBAT	Module extended power supply voltage (2)	The actual input voltage must be	3.1	3.8	4.5	V

		within this range.				
I _V BAT	Peak current (per Launch time slot below)	Under the maximum transmission power level	-	1.0	1.2	A
USB_VBUS	USB detection	-	3.0	5.0	5.25	V
USIM_VDD	USIM card power supply output	-	-	1.8/3.0	-	V
VDD_EXT	Reference power output	-	-	1.8	-	V

Precautions

◆ When the module operates within this voltage range, its related performance meets the 3GPP standard requirements.

◆ When the module operates within this voltage range, it can still maintain normal working status without irreversible faults; only individual indicators, such as output power and other parameter values, may exceed the 3GPP standard range. When the voltage returns to the normal operating range, all indicators of the module still comply with 3GPP standards.

◆ Sleep mode: VDD_EXT does not lose power in sleep mode.

7.3 Power consumption

The power consumption of the module is as follows:

Table 6-3 Module Power Consumption

Module description	Test conditions	Typical value	Unit
Shutdown mode	Module shutdown	0.86	μ A
Sleep mode	AT+CFUN=0 (USB disconnected)	3.1	μ A
	LTE-FDD @ PF = 32 (USB disconnected)	0.63	mA

	LTE-FDD @ PF = 64 (USB disconnected)	0.34	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.17	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.15	mA
	LTE-TDD @ PF = 32 (USB disconnected)	0.69	mA
	LTE-TDD @ PF = 64 (USB disconnected)	0.37	mA
	LTE-TDD @ PF = 128 (USB disconnected)	0.18	mA
	LTE-TDD @ PF = 256 (USB disconnected)	0.13	mA
Idle mode	LTE-FDD @ PF = 64 (USB disconnected)	4.1	mA
	LTE-FDD @ PF = 64 (USB Link)	27.8	mA
	LTE-TDD @ PF = 64 (USB disconnected)	4.1	mA
	LTE-TDD @ PF = 64 (USB Link)	27.8	mA
LTE data transmission	LTE-FDD B1	619	mA
	LTE-FDD B3	584	mA
	LTE-FDD B5	546	mA
	LTE-FDD B8	531	mA
	LTE-TDD B34	200	mA
	LTE-TDD B38	202	mA
	LTE-TDD B39	186	mA
	LTE-TDD B40	207	mA
	LTE-TDD B41	204	mA

7.4 Digital logic level characteristics

The logic level definition of the module digital IO is as follows:

Table 6-4 Digital IO Logic Level Description

Type	Parameter	Description	Minimum value	Maximum value	Unit
Input	VIL	Input low level	-0.3	0.6	V

	V _{IH}	Input high level	1.2	2	V
Output	V _{OL}	Output low level	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Precautions

The GPIO reference level of the module follows VDD_EXT, and VDD_EXT will lose power during sleep, so the corresponding GPIO will also...
It will lose power.

The AGPIO logic level definition of the NT26-KCN E module is as follows (AON_VDD range: 1.7~1.9V; typical value: 1.8V):

Table 6-5 Digital IO Logic Level Description

Type	Parameter	Description	Minimum value	Maximum value	Unit
Input	V _{IL}	Input low level	-0.3	0.6	V
	V _{IH}	Input high level	1.2	2	V
Output	V _{OL}	Output low level	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Precautions

The AGPIO reference level of the module follows LDO_AONIO and will not lose power during sleep.

The logic level definition of the module USIM card interface is as follows:

Table 6-6 USIM 1.8V Logic Level Description

Type	Parameter	Description	Minimum	Maximum	Unit
------	-----------	-------------	---------	---------	------

			value	value	
Input	V_{IL}	Input low level	-	0.6	V
	V_{IH}	Input high level.	1.2	-	V
Output	V_{OL}	Output low level	-	0.45	V
	V_{OH}	Output high level	1.35	-	V

Table 6-7 USIM 3.0V Logic Level Description

Type	Parameter s	Description	Minimum value	Maximum value	Unit
Input	V_{IL}	Input low level	-	1.0	V
	V_{IH}	Input high level.	1.95	-	V
Output	V_{OL}	Output low level	-	0.45	V
	V_{OH}	Output high level	2.55	-	V

Precautions

The level of USIM_VDD will be determined based on the detected category of the USIM card, supporting 1.8V/3.0V USIM cards.

7.5 Electrostatic Protection

In daily life and production, static electricity is everywhere. Static electricity from the human body or generated by the friction of objects can be transmitted to the module through various means and may cause damage to the module. Therefore, it is essential to pay close attention to static electricity protection and take necessary protective measures. For example:

◆ Wear anti-static gloves during the processes of research and development, production, assembly, and testing.

◆ When designing products, add anti-static protection devices at circuit

interfaces and other points prone to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-8 Electrostatic Protection Characteristics

Test interface	Contact discharge	Remarks
VBAT	$\pm 8\text{KV}$	Test Standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	$\pm 8\text{KV}$	
Antenna interface	$\pm 8\text{KV}$	
Others	-	

The detailed design of electrostatic protection refers to the "Lierda NT26-KCN E Series Hardware Reference Design Manual."

7.6 Working and storage temperature

Table 6-9 Operating Temperature Range

Parameter	Minimum value	Typical value	Maximum value	Unit	Remarks
Normal operating temperature(1)	-35	+25	+75	°C	
Extended operating temperature(2)	-40	+25	+85	°C	
Storage environment temperature(3)	-40	+25	+90	°C	

Precautions

(1) When the module operates within this temperature range, its relevant performance meets the 3GPP standard requirements.

(2) When the module operates within this temperature range, it can still maintain normal working status without experiencing unrecoverable failures; only certain indicators, such as output power and other parameter values, may exceed the 3GPP standard range. When the temperature returns to the normal operating range, all indicators of the module

still comply with 3GPP standards.

(3) This storage temperature range does not include packaging materials, and attention should be paid to the maximum tolerance temperature of the tape packaging.

7.7 Precautions

There are some precautions to be taken during the production and use of the module:

- ◆ When spraying the module, please try to avoid the spraying material flowing into the interior of the module.
- ◆ When cleaning the module, do not perform ultrasonic cleaning on the module, as this may cause damage to the internal crystals of the module.
- ◆ During the production and use of the module, please avoid applying it in environments or enclosures containing any amount of mercury or mercury vapor, as this may lead to the risk of product failure or malfunction.

8 Mechanical dimensions

8.1 Mechanical dimensions

The mechanical dimensions of the module are as follows:

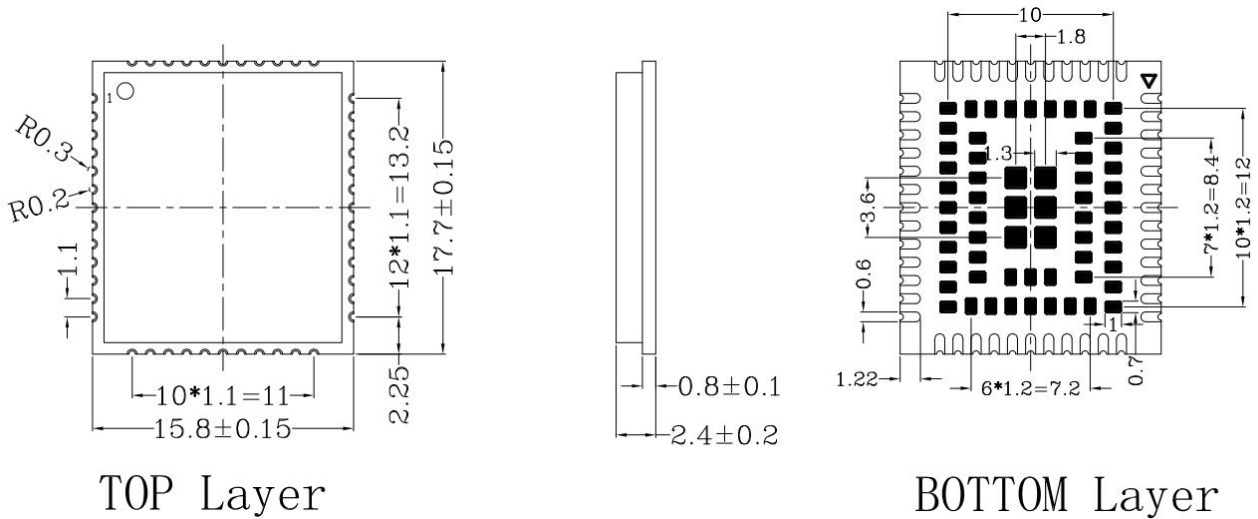


Figure 7.1 Module Mechanical Dimensions (Unit: mm)

8.2 Top view/bottom view of the module

The physical effect diagram of the module is as follows, subject to the actual module.



Figure 7.2 Module Top View/Bottom View

8.3 Recommend packaging

The recommended module pads are shown in the figure below, and users can make slight adjustments based on their own production processes.

◆ The module's surrounding pins are designed with rounded corners internally. When designing the baseboard pads, please consider using rounded corner transitions. For the rectangular pads at the bottom of the module, the baseboard design can adopt the module pin dimensions, as shown in the reference design diagram of a single pad below.

◆ In order to facilitate the opening of stepped stencils, it is recommended not to place other components within 2.0mm outside the module pads. This distance can be determined by the user based on the requirements of their stencil manufacturer.

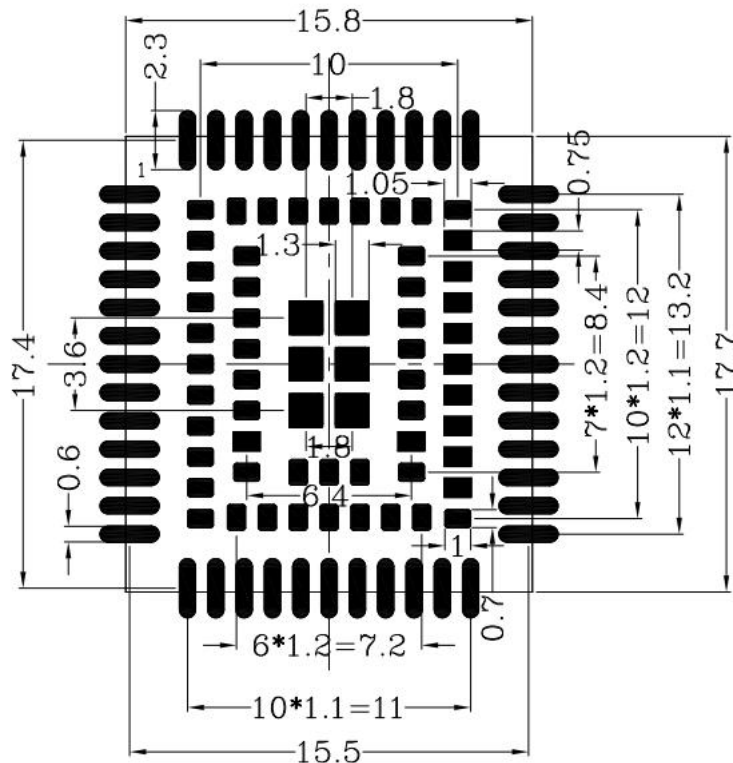


Figure 7.3 Module Recommended Pad (Unit: mm)

9 Production and Packaging Information

This chapter describes the module's surface-mount technology, packaging methods, storage conditions, and other information, which can help users better preserve and use this module.

9.1 Production welding

9.1.1 Reflow method

When mounting the module, the following points need to be noted during reflow soldering:

- ◆ To ensure the welding quality of the module, the recommended thickness of the stencil corresponding to the module pad section is 0.18-0.2mm;
- ◆ If the baseboard using the module is double-sided SMT, it is recommended to place the module during the second SMT process to avoid damage to the module caused by repeated heating.
- ◆ During the first SMT process, it is best for the customer's baseboard to pass through the reflow oven on the conveyor belt, and during the second SMT process, it should also preferably pass through the reflow oven on the conveyor belt.
- ◆ If, due to special reasons, it cannot be placed on the mesh belt for reflow, consider using a fixture to pass through the reflow on the track or placing a flat, high-temperature-resistant straight template to support the PCBA during reflow, preventing PCB deformation during reflow that could lead to module soldering defects.

9.1.2 Reflow Soldering Operation Guide

The PCBA reflow oven temperature profile is related to the solder paste used and needs to be adjusted according to the actual solder paste. The data is only suitable for

lead-free operations. Refer to Figure 8-2 Lead-Free Reflow Soldering Operation Guide.

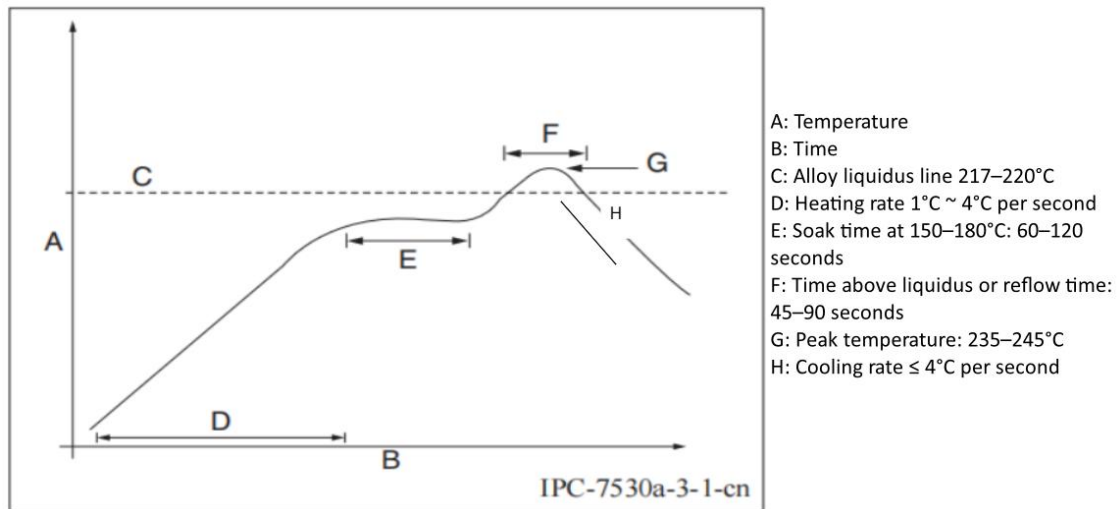


Figure 8.1 Lead-Free Reflow Soldering Work Instructions

Remarks

(1) During the production process of welding or other activities that may directly contact the module, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) should be used to wipe the module shield; otherwise, it may cause the shield to rust.

(2) The module's nickel silver laser-engraved shielding cover can meet the following: after a 12-hour neutral salt spray test, the laser-engraved information remains clear and identifiable, and the QR code can be scanned (there may be white rust).

(3) If the module needs to be sprayed, please ensure that the spraying material does not chemically react with the module's shield or PCB, and also ensure that the spraying material does not flow into the module.

(4) Do not perform ultrasonic cleaning on the module, as it may cause damage to the internal crystal of the module.

(5) Due to the complexity of the SMT process, in case of uncertain situations or processes not mentioned in the documentation (such as selective wave soldering or ultrasonic welding), please confirm with Lierda technical support before starting the SMT process.

9.1.3 Maintenance

If the module has issues such as cold solder joints or short circuits and needs repair, please proceed according to the following parameters:

◆ Lead-free process: Soldering iron temperature $380 \pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$.

◆ Leaded process: Soldering iron temperature $350 \pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$.

9.2 Packaging Specifications

The module is packaged using a plastic wheel carrier tape method upon delivery, with the reference dimensions of the plastic wheel as follows:

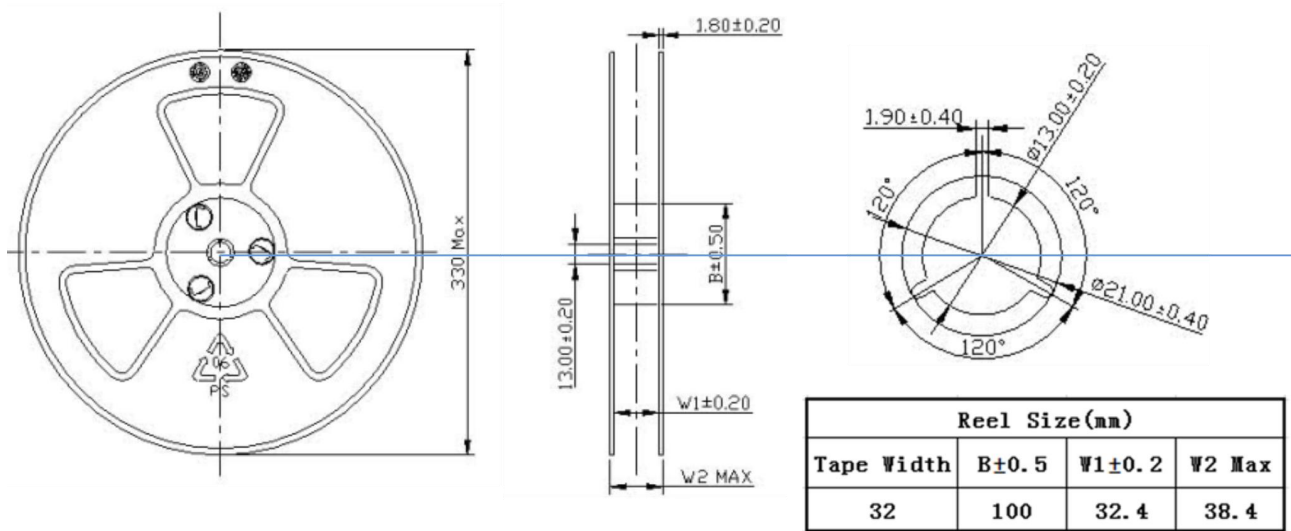


Figure 8.2 Carrier Tape Rubber Wheel Dimension Diagram

The tape feeding direction is as follows: (Refer to the diagram, label content is subject

to the actual product, pay attention to the PIN1 position of the module).

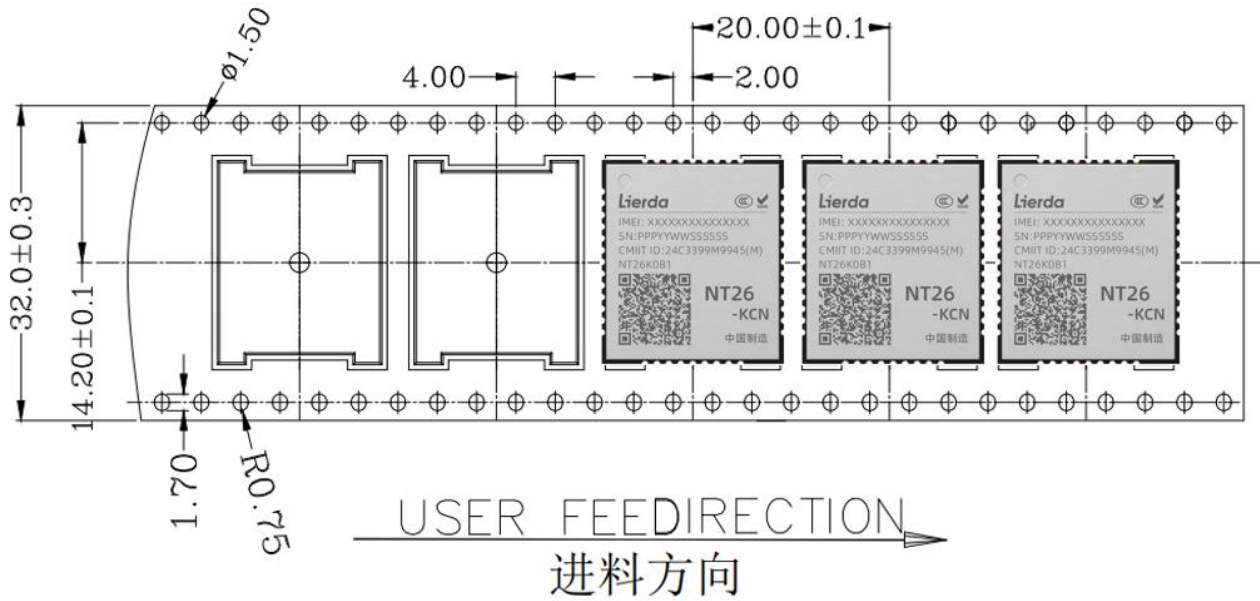


Figure 8.3 Carrier Tape Feeding Direction Indication Diagram

9.3 Storage conditions

The module is shipped in the form of a vacuum-sealed bag. The module's moisture sensitivity level is 3 (MSL 3), and its storage must comply with the following conditions:

1. Recommended storage conditions: temperature 23 ± 5 °C, and relative humidity 35~60%.
2. Under the recommended storage conditions, the module can be stored in a vacuum-sealed bag for 12 months.
3. Under workshop conditions with a temperature of 23 ± 5 °C and relative humidity below 60%, the workshop lifespan of the module after unsealing is 168 hours. Under these conditions, the module can be directly subjected to reflow production or other high-temperature operations. Otherwise, the module needs to be stored in an environment with relative humidity less than 10% (e.g., a moisture-proof cabinet) to maintain the module's dryness.
4. If the module is under the following conditions, pre-baking treatment is required to prevent the module from absorbing moisture, which may lead to PCB bubbling, cracking,

and delamination after high-temperature soldering:

- Storage temperature and humidity do not meet the recommended storage conditions;
- After the module is unsealed, production or storage is not completed according to item 3 above;
- Vacuum packaging leaks, material in bulk;
- Before module repair.

5. Baking process of the module:

- Needs to be baked at high temperature for 8 hours under the condition of $120 \pm 5^{\circ}\text{C}$;
- Modules that undergo secondary baking must be soldered within 24 hours after baking; otherwise, they must still be stored in a drying box.

Precautions

(1)To prevent and reduce welding defects such as bubbling and delamination caused by moisture in the module, strict control should be implemented, and it should not be exposed to air for a long time after the vacuum package is opened.

(2)Before baking, remove the original packaging and transfer the module to a high-temperature-resistant carrier to prevent plastic trays or reels from heat deformation. If only short-term baking is required, follow the IPC/JEDEC J-STD-033 standard.

(3)When unpacking and transferring modules, anti-static measures must be taken, such as wearing anti-static gloves.

10 Relevant documents and terminology abbreviations

10.1 Relevant documents

The relevant document reference is as follows:

Table 9-1 Related Documents

Serial Number	Document Name
[1]	Lierda NT26-KCN E Series Hardware Reference Design Manual
[2]	Lierda LTE Standard(L) Series AT Command Manual
[3]	Lierda LTE Standard (Q) Series AT Command Manual
[4]	IPC/JEDEC J-STD-033 Standard

10.2 Abbreviation of terms

The abbreviations and their related definitions involved in this document are as follows:

Table 9-2 Abbreviation of Terms

Abbreviation	Full English name	Full name in Chinese.
3GPP	3rd Generation Partnership Project	The Third Generation Partnership Program
ADC	Analog-to-Digital Converter	Analog-to-Digital Conversion
ANT	Antenna	Antenna
ASM	Antenna Switch Module	Antenna switch module
DAC	Digital -to- Analog Converter	Digital-to-Analog Conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
DCXO	Digitally Controlled Crystal Oscillator	Digital Controlled Crystal Oscillator

DRX	Discontinuous Reception	Non-continuous reception
DTE	Data Terminal Equipment	Data terminal equipment
ECL	Equivalent Class Level	Network coverage level
ESD	Electro-Static discharge	Electrostatic discharge
EOS	Electrical Overtress	Electrical Overstress (Surge)
ESR	Equivalent Series Resistance	Equivalent series resistance
EVK	Eval μ Ation Kit	Evaluation toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half-Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	General input and output
I/O	Input/Output	Input and output interface
I _{max}	Maximum Load Current	Maximum current
I _{norm}	Normal Current	Normal (Typical) Current
bps	Bits Per Second	Rate unit
LCC	Leadless Chip Carriers	Leadless chip carrier package
LCM	LCD Module	LCD display module
LDO	Low Dropout Regulator	Low Dropout Linear Regulator
LGA	Land Grid Array	Grid Array Package
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M Protocol
MCU	Mirco Controller Unit	Microcontroller Unit
MSL	Moisture Senticity levels	Moisture Sensitivity Level
PCB	Printed Circuit Board	Printed circuit board
PCBA	Printed Circuit Board Assembly	Printed Circuit Board Assembly
PMU	Power Management Unit	Power Management Unit

PSM	Power Saving Mode	Energy-saving mode
RF	Radio Frequency	Radio Frequency
RoHS	Restriction of Hazardous Substances	Restriction of Hazardous Substances
RX	Receive	Receive
TAU	Tracking Area Update	Track area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient Suppression Diode
TX	Transmit	Send
μ ART	Universal Asynchronous Receiver & Transmitter	Universal Asynchronous Receiver and Transmitter
UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
URC	Unsolicited Result Code	Non-request result code
(U)SIM	(Universal) Subscriber Identification Module	(Generic) User Identity Recognition Module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (Typical) Voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level
VImax	Absolute Maximum Input Voltage Value	Maximum input level
VImin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level

VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level